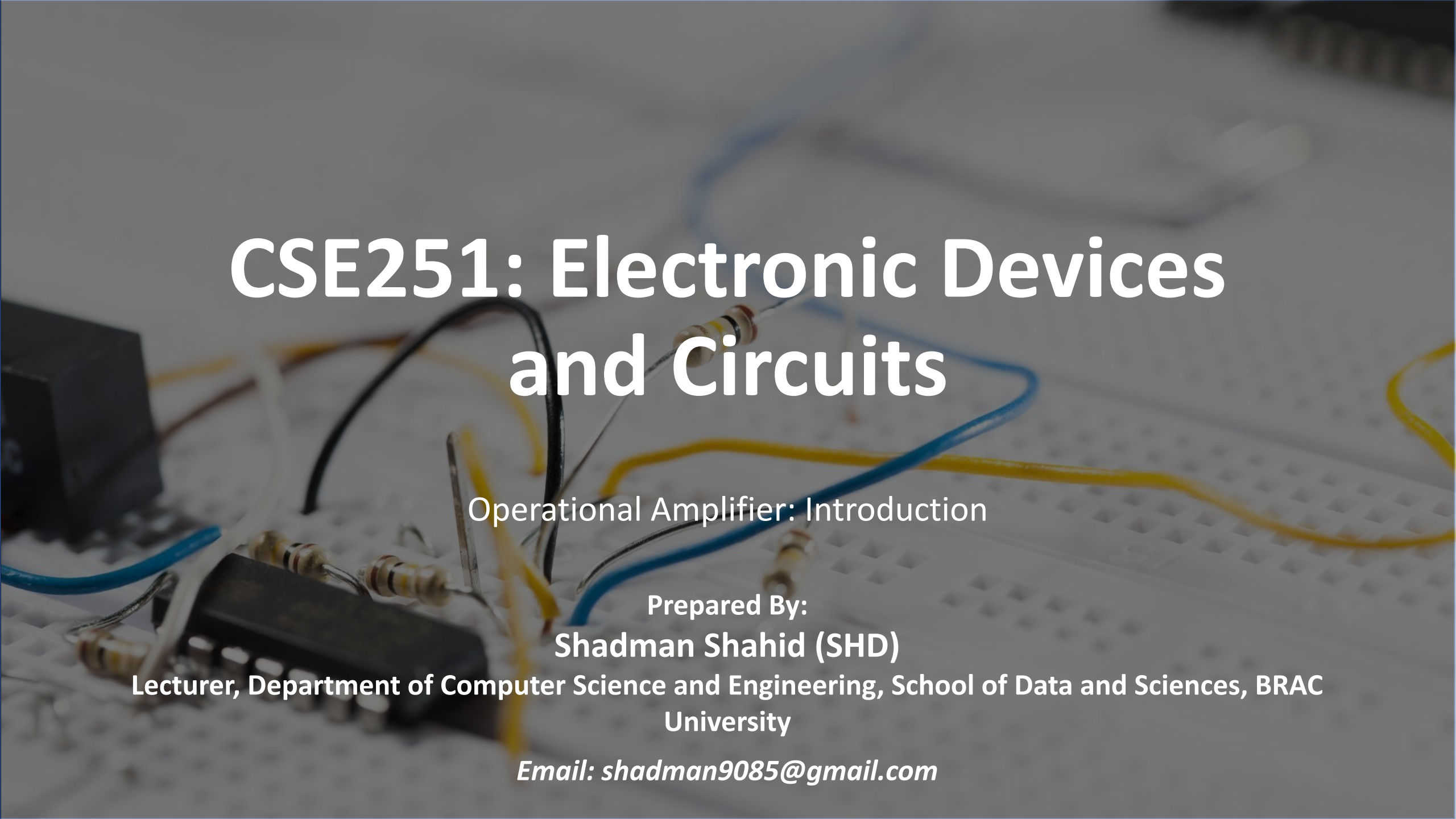




CSE251: Electronic Devices and Circuits

Operational Amplifier: Introduction

Prepared By:

Shadman Shahid (SHD)

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University

Email: shadman9085@gmail.com

Outline

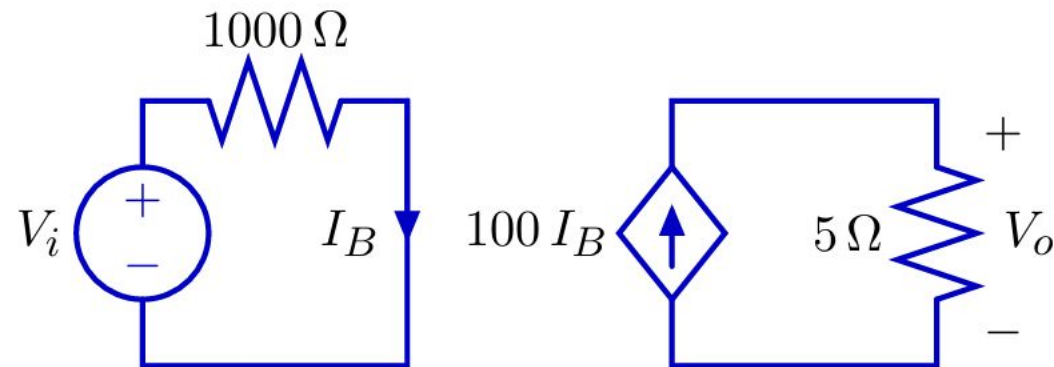
- Dependent Sources
- Operational Amplifier: Introduction
- Op-Amp: Circuit Symbols and terminal
- Op-Amp: VTC (Voltage Transfer Characteristics)
 - Linear Amplification
 - (Positive and Negative) Saturation
- Op-Amp: Examples
- Op-Amp: Physical Entity

Dependent Sources

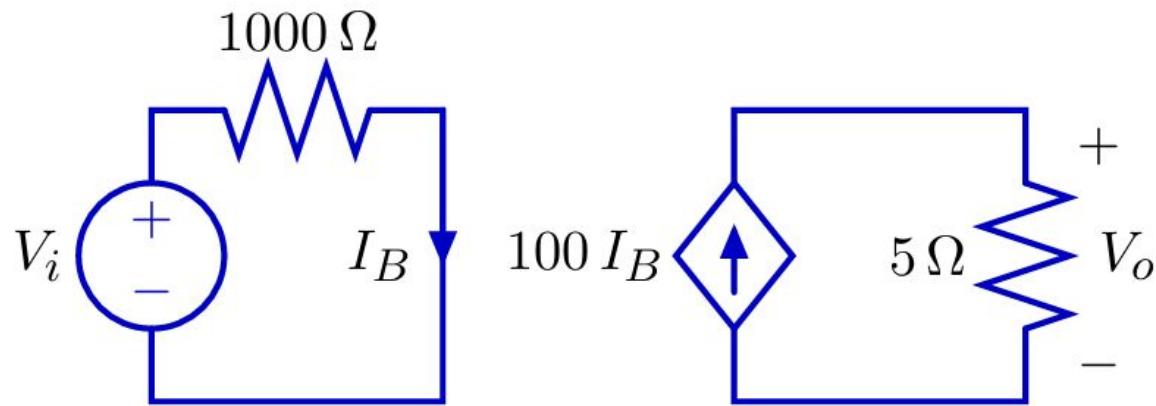
To analyze op-amps, we must understand **dependent source**.

A dependent source generates a voltage or current whose value depends on another voltage or current.

Example: current-controlled current source



Dependent Sources



$$I_B = \frac{V_i}{1000\ \Omega}$$

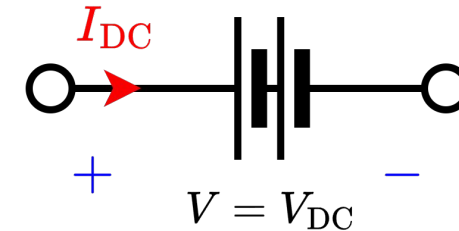
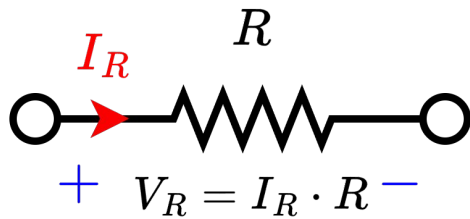
$$V_o = 100 I_B \times 5\ \Omega$$

$$\text{Voltage Gain: } \frac{V_o}{V_i} = \frac{1}{2}$$

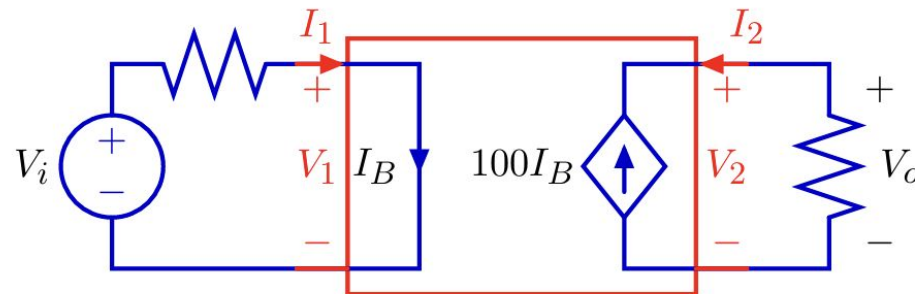
$$= 100 \frac{V_i}{1000\ \Omega} \times 5\ \Omega = \frac{1}{2} V_i$$

Dependent Sources

Resistors, (Ind.) Voltage sources, (Ind.) Current sources are single “**port**” device. They are characterized by a **single equation**.



Dependent sources are two-ports: characterized by **two** equations.



Here $V_1 = 0$ and $I_2 = -100 I_1$.

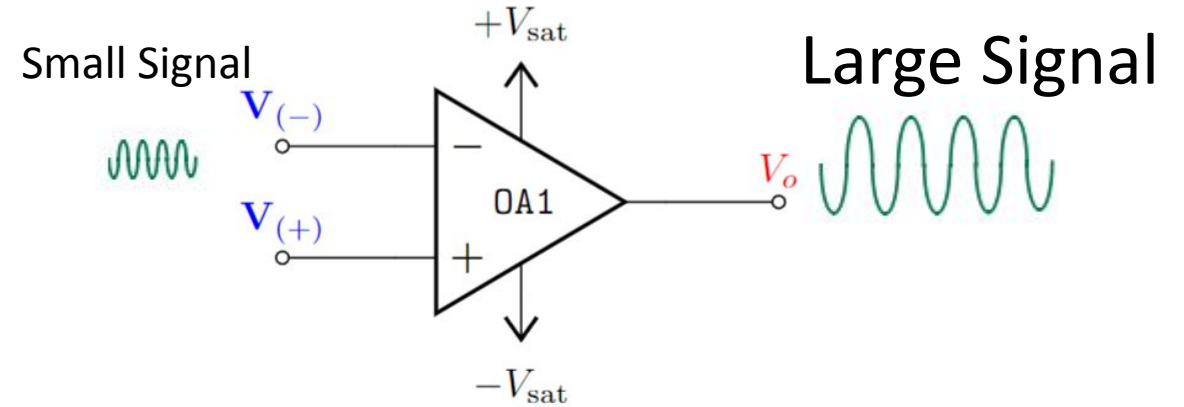
Operational Amplifier: Introduction

- **Operational:**

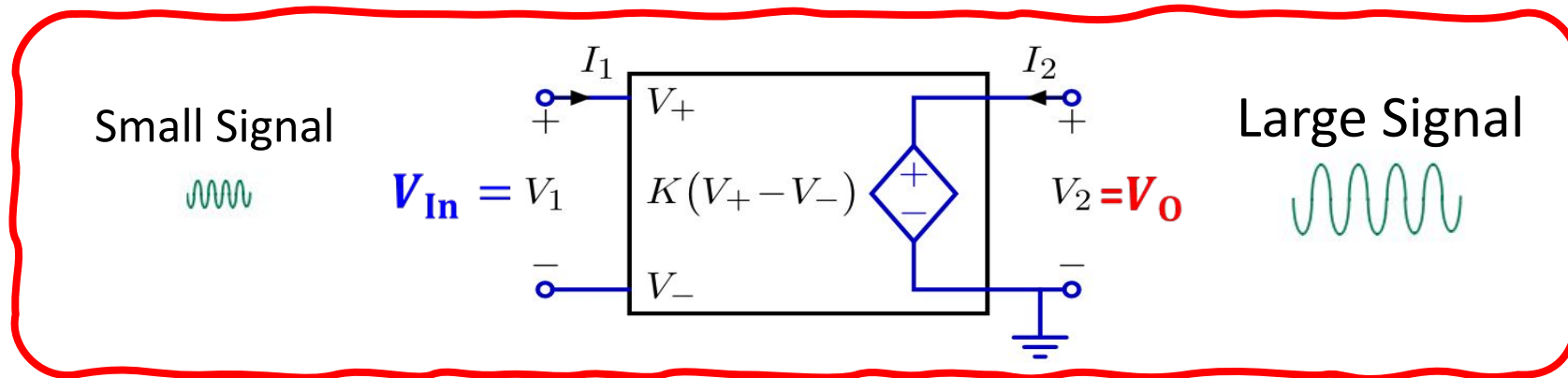
Mathematical Operations

- **Amplifier:**

Amplifies input signal/voltage.

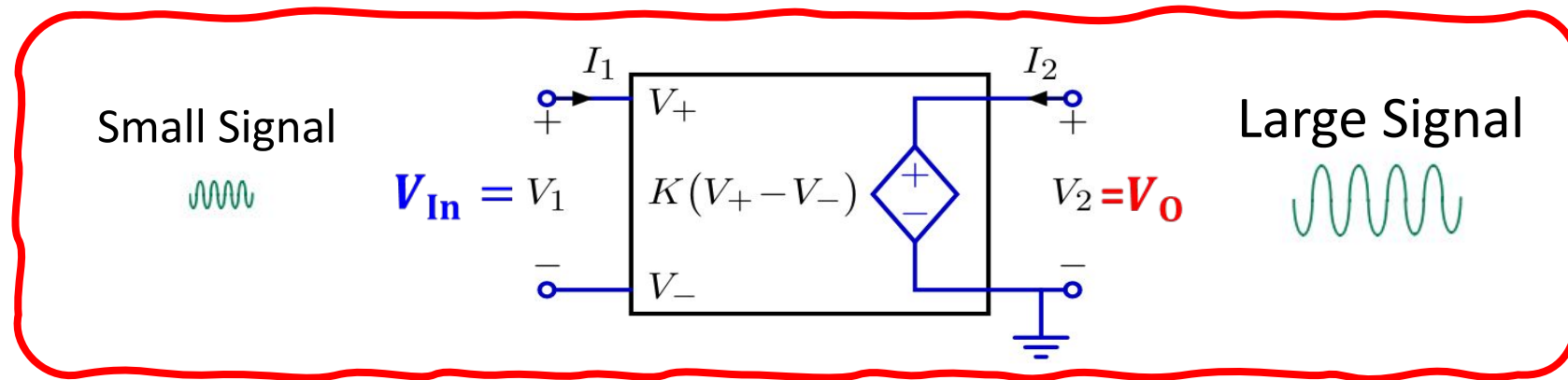


An **op-amp (operational amplifier)** can be modelled by a **voltage- controlled voltage source**.



Operational Amplifier: Introduction

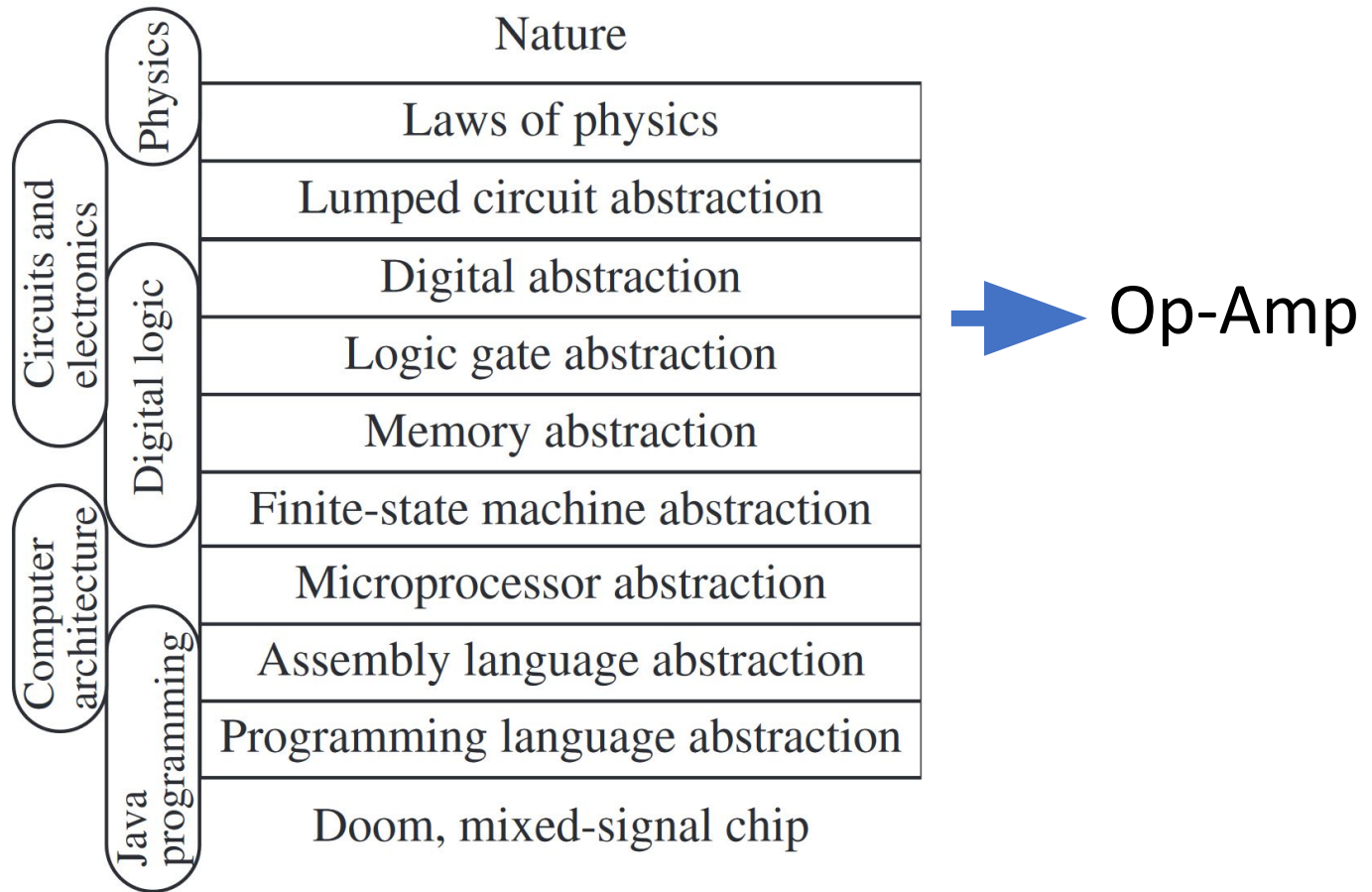
An **op-amp** (operational amplifier) can be represented by a **voltage- controlled voltage source**.



$I_1 = 0$ and $V_O = K V_{In}$ where K is large (typically $K > 10^5$).

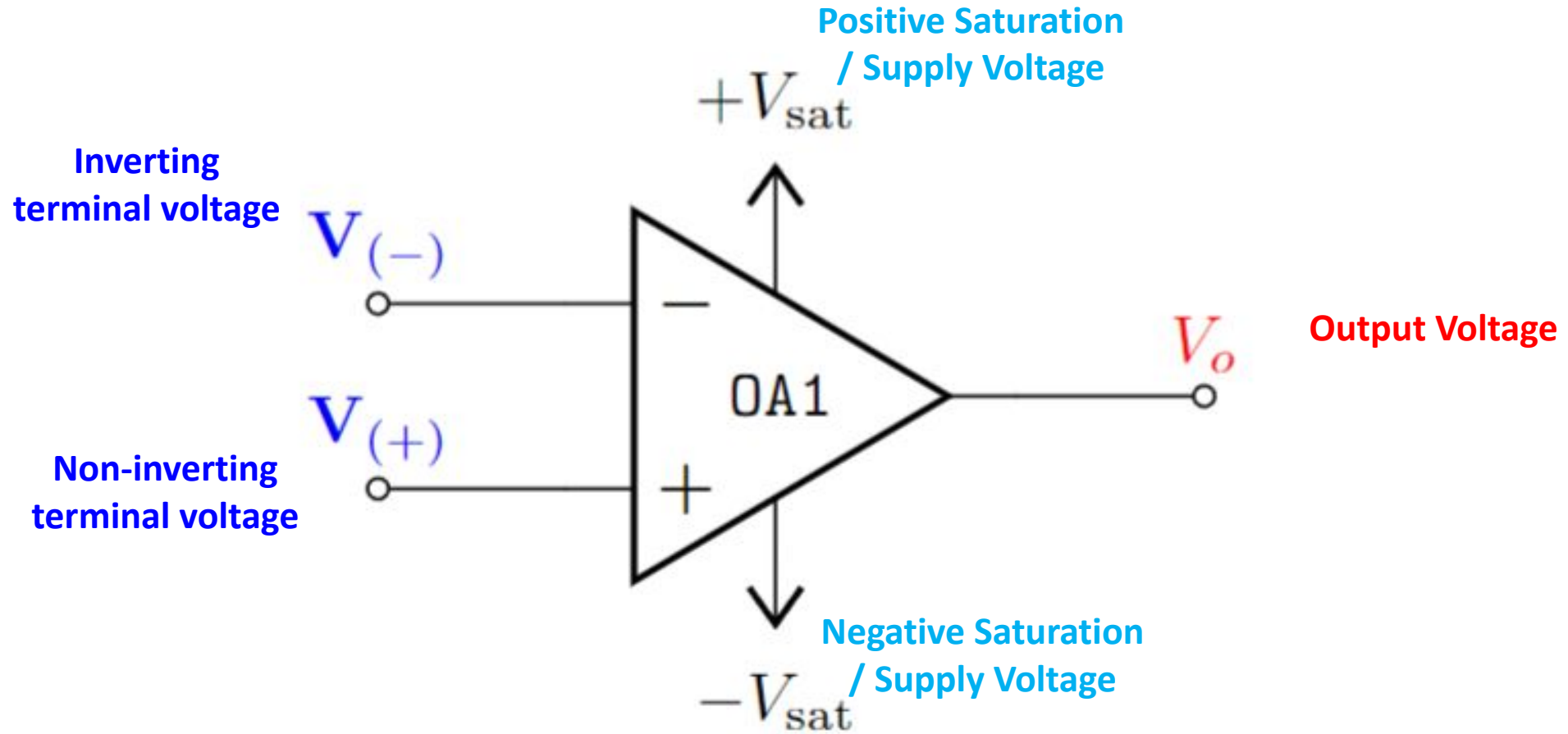
Operational Amplifier: Introduction

The Circuit Abstraction

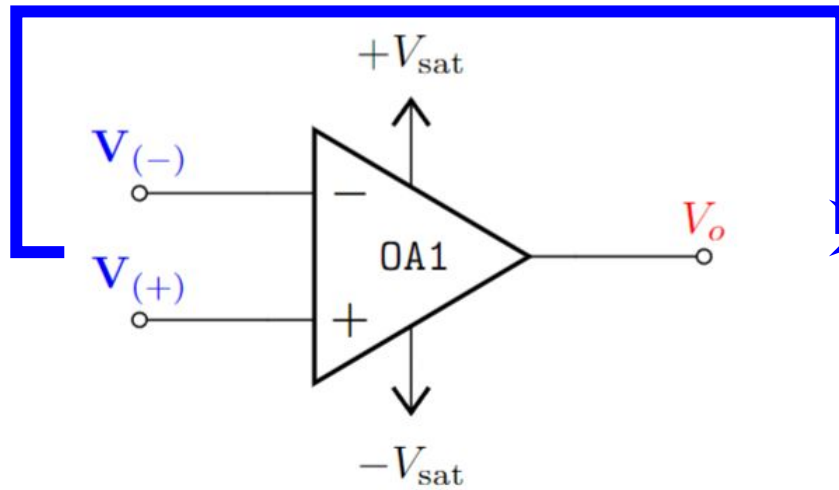


Sequence of courses and the abstraction layers

Op-Amp: Circuit Symbols and terminal



Op-Amp: Circuit Symbols and terminal

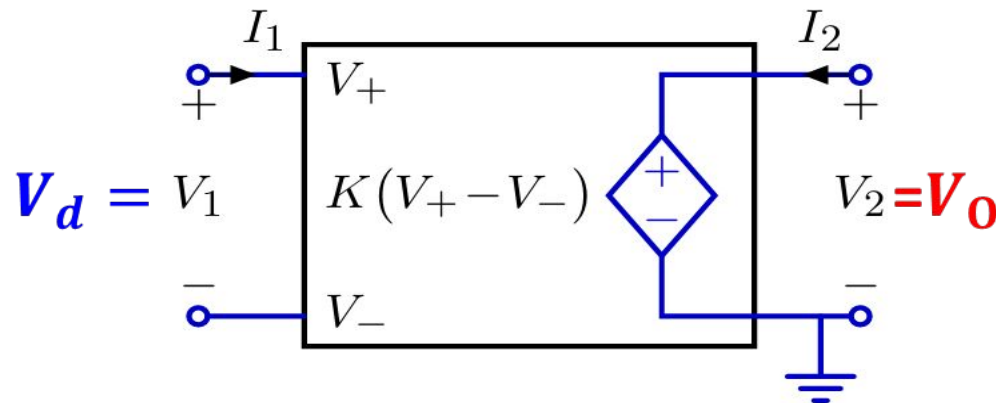


$$V_d = V_{(+)} - V_{(-)}$$

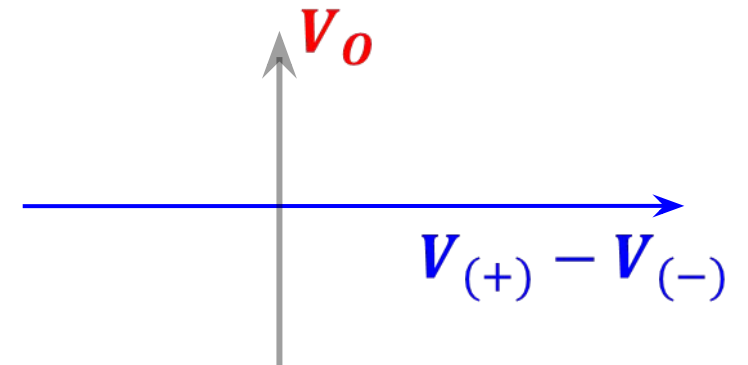
Difference Amplifier –

Amplifies the voltage difference between two terminals.

Differential input voltage



Voltage Transfer Characteristics (VTC)



Op-Amp: VTC – Linear Amplification

Voltage Transfer Characteristics (VTC)

- Shows how the **output voltage** varies with the **input voltage** $V_O(V_d)$

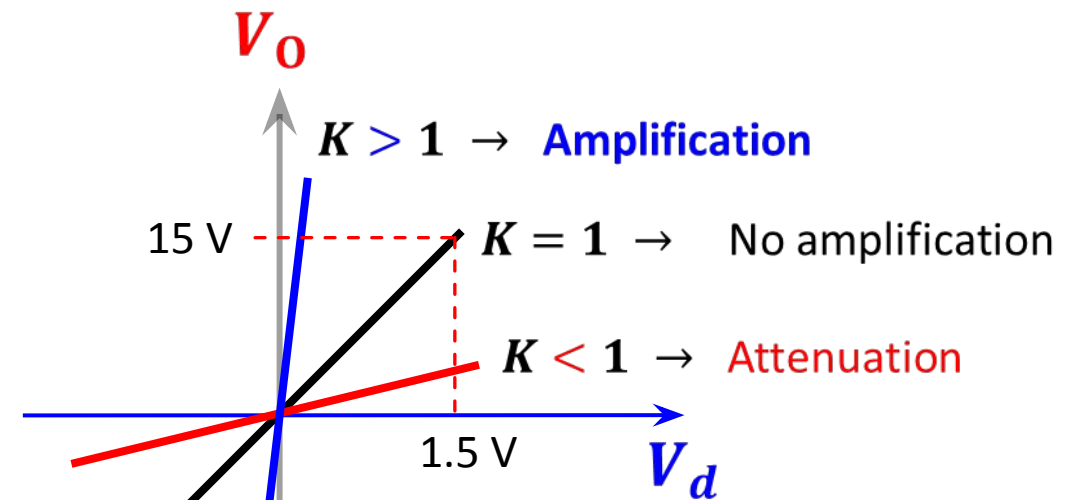
- y – axis $\rightarrow V_O$

- x – axis $\rightarrow V_d$

- Slope: $K = \frac{\Delta V_O}{\Delta V_d} = \frac{V_O}{V_d}$

If the line passes through origin:

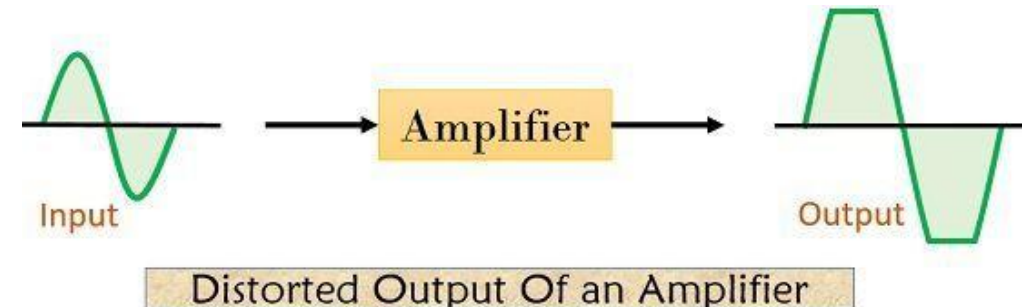
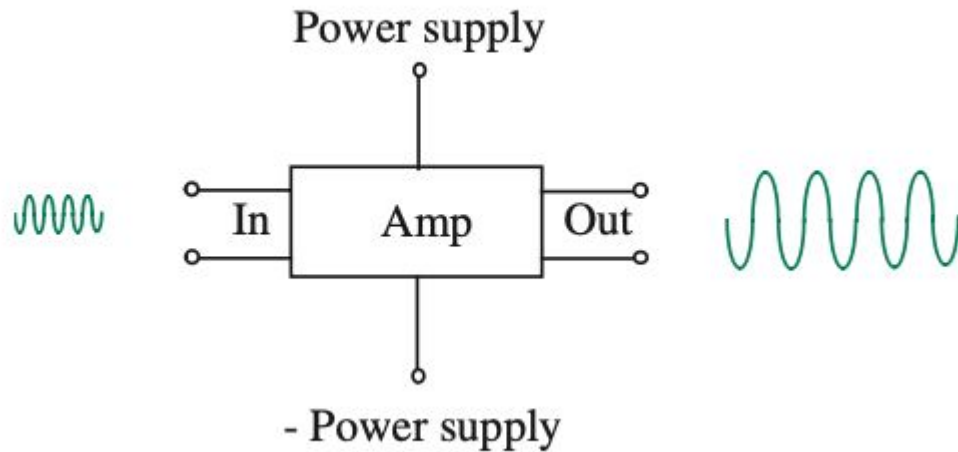
$$y = K \cdot x$$
$$V_O = K \cdot V_d$$



LINEAR AMPLIFICATION

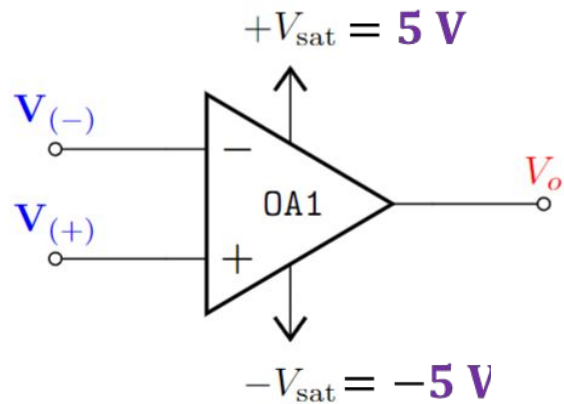
Op-Amp: VTC – Linear Amplification

- **Linear Amplification** only takes place within a [valid input range](#).
- Otherwise output will be distorted - - Saturation



The limiting factor of **linear amplification** is determined by the **power supply** to the amplifier

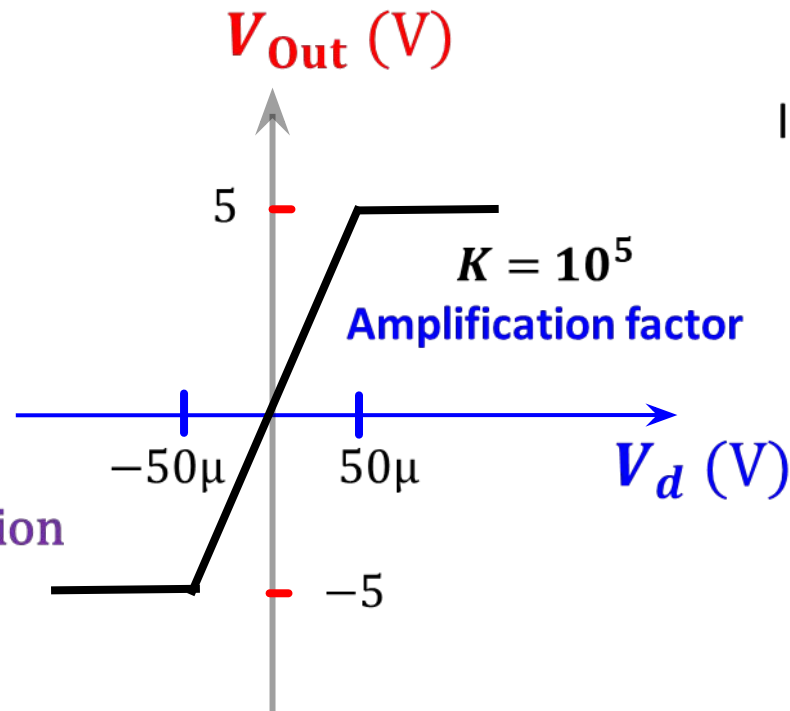
Op-Amp: VTC - Saturation



$$V_o = K \cdot V_{In} : \text{When } -5\text{ V} < V_o < 5\text{ V}$$

$K \rightarrow 10^5$: Gain / Amplification

Non-Linear characteristics



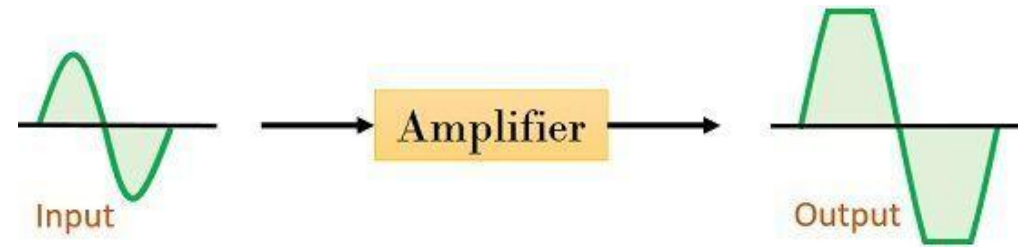
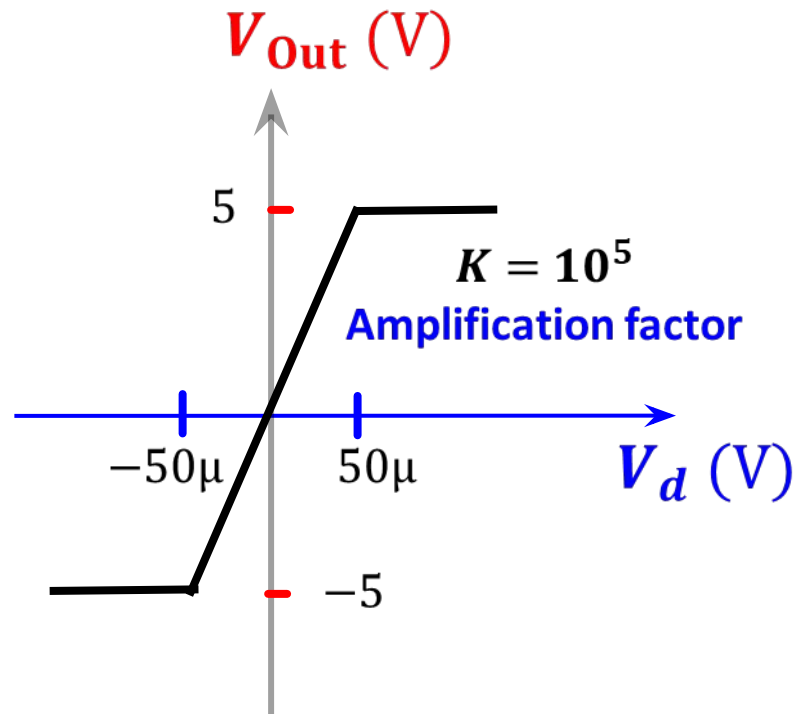
If $V_d > 50\mu$: Positive Saturation

$$\Rightarrow V_o = 5$$

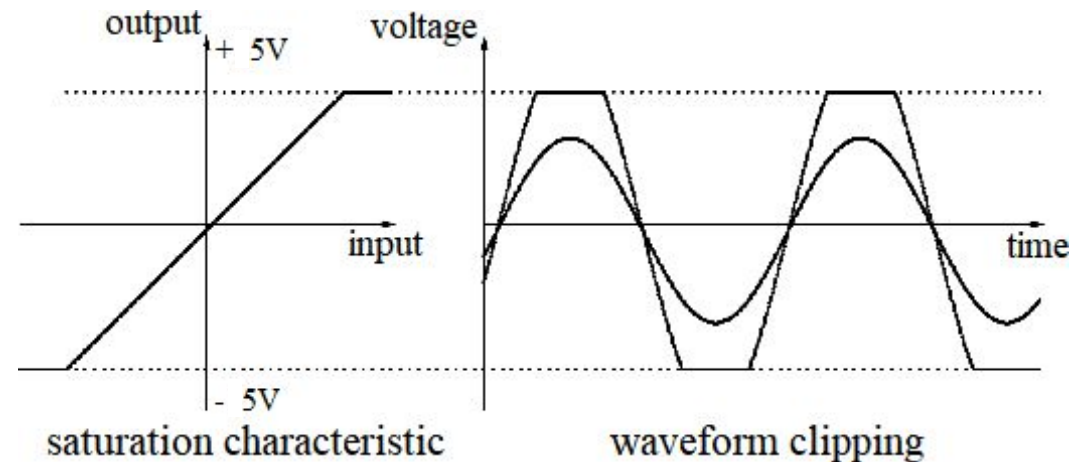
If $V_d < -50\mu$: Negative Saturation

$$\Rightarrow V_o = -5$$

Op-Amp: VTC - Saturation



Distorted Output Of an Amplifier



Op-Amp: VTC - Summary

Voltage Transfer Characteristics (VTC)

Positive saturation:

If $V_d > \frac{+V_{\text{sat}}}{A}$: Positive

$$\Rightarrow V_o = +V_{\text{sat}}$$

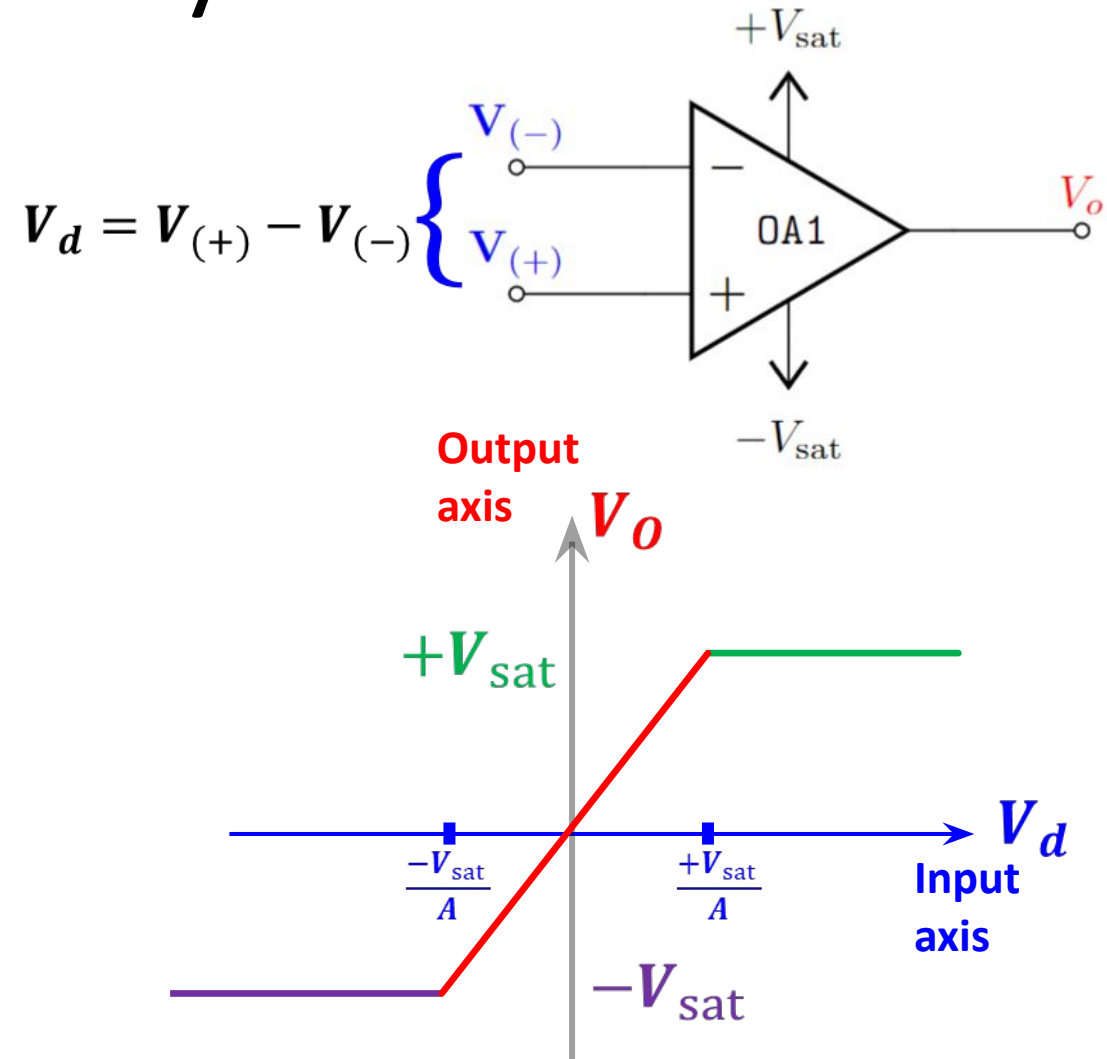
Linear Region

$V_o = AV_d$: When V_d is very small
 $-V_{\text{sat}} < V_o = AV_d < +V_{\text{sat}}$

Negative saturation:

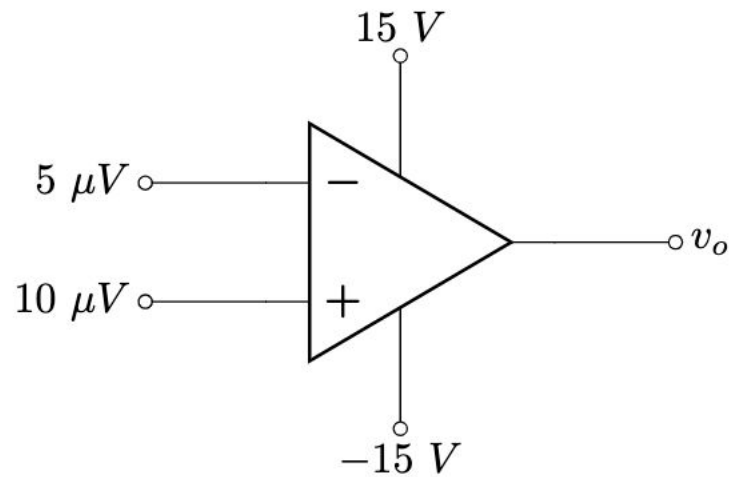
If $V_d < \frac{-V_{\text{sat}}}{A}$: Negative

$$\Rightarrow V_o = -V_{\text{sat}}$$

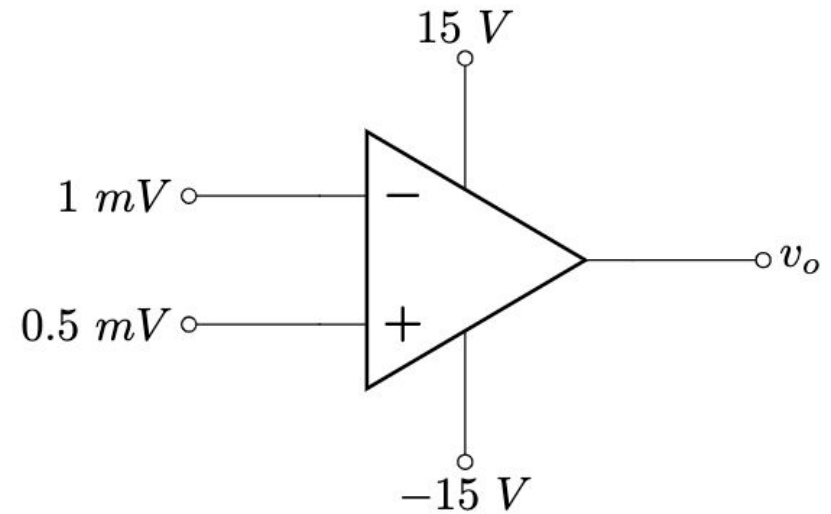


Op-Amp: Examples

Find v_o



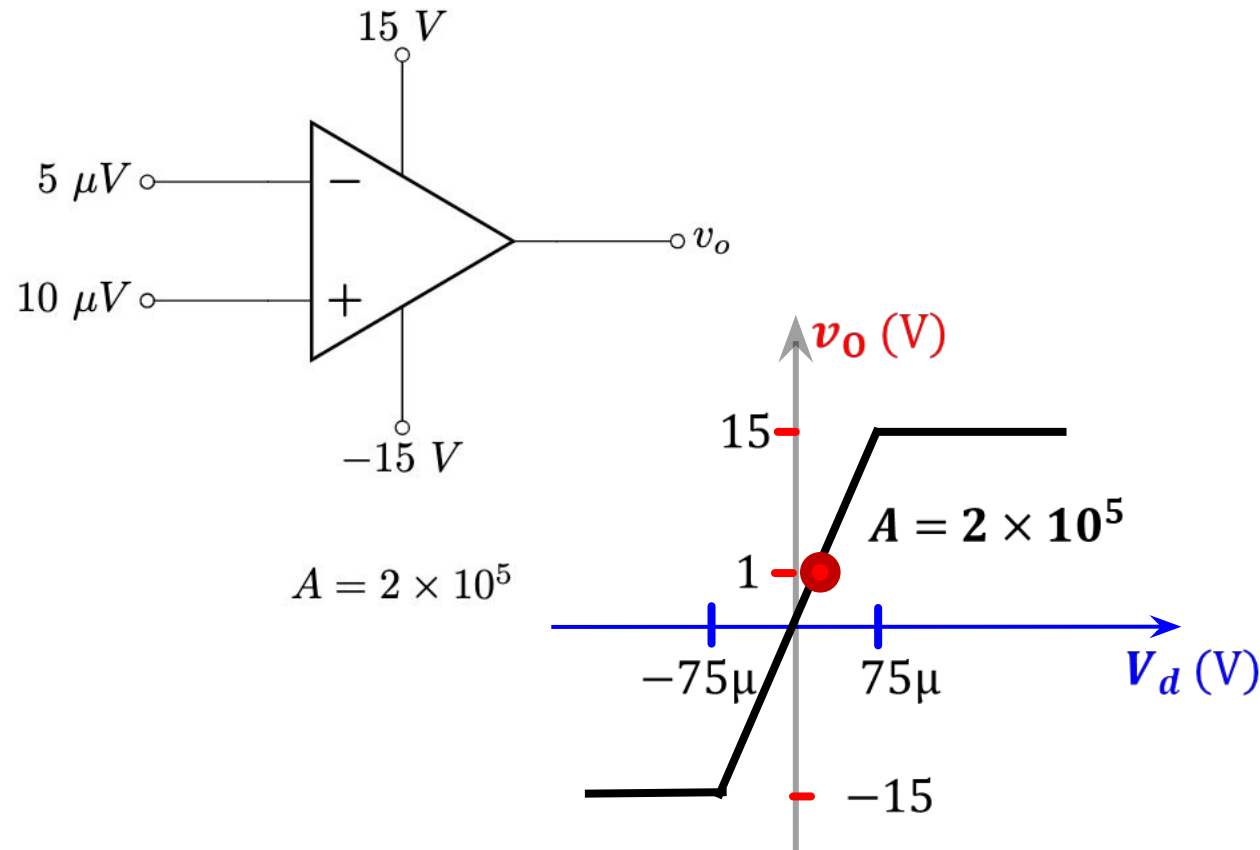
$$A = 2 \times 10^5$$



$$A = 2 \times 10^5$$

Example 1

- Find v_o



Solution:

$$V_d = V_{(+)} - V_{(-)} = (10 - 5)\ \mu\text{V} = 5\ \mu\text{V}$$

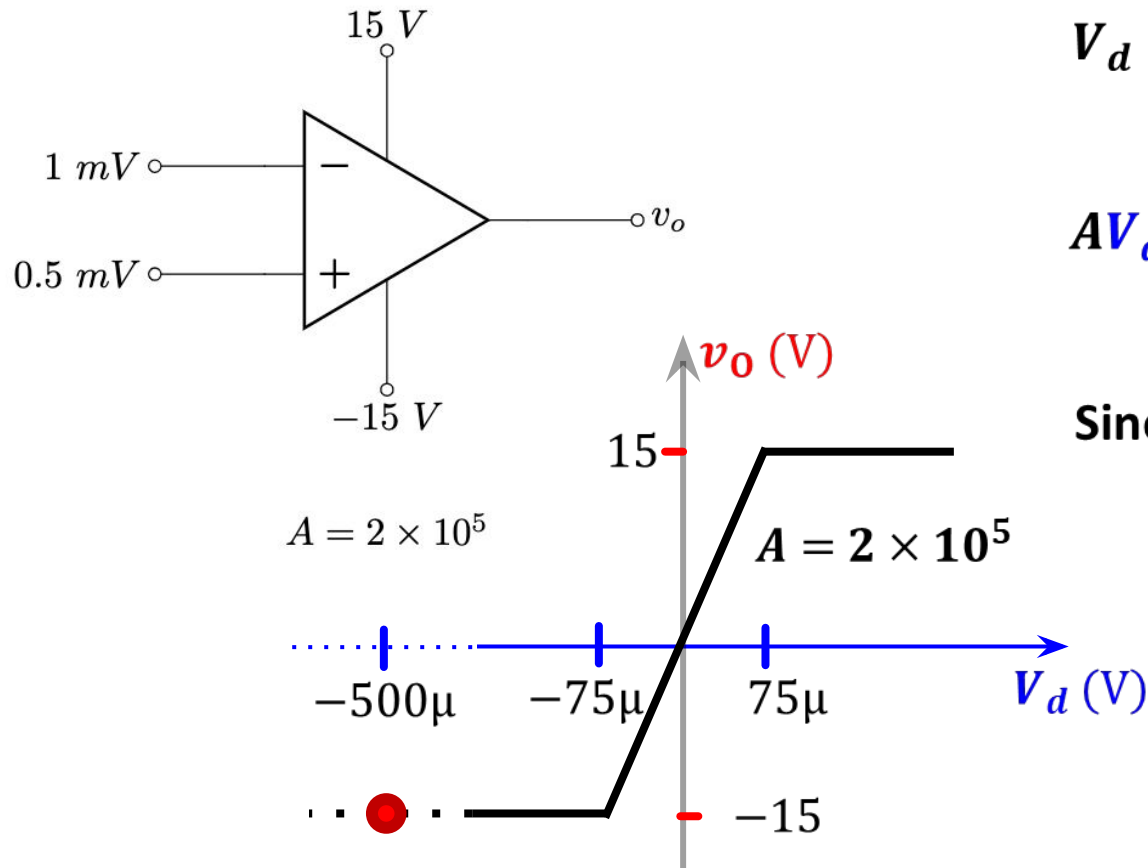
$$AV_d = (2 \times 10^5) \times (5 \times 10^{-6})\ \text{V} = 1\ \text{V}$$

Since $-15\ \text{V} < AV_d < +15\ \text{V}$

$$V_o = AV_d = 1\ \text{V}$$

Example 2

- Find v_o



Solution:

$$V_d = V_{(+)} - V_{(-)} = (0.5 - 1) \text{ mV} = -0.5 \text{ mV}$$

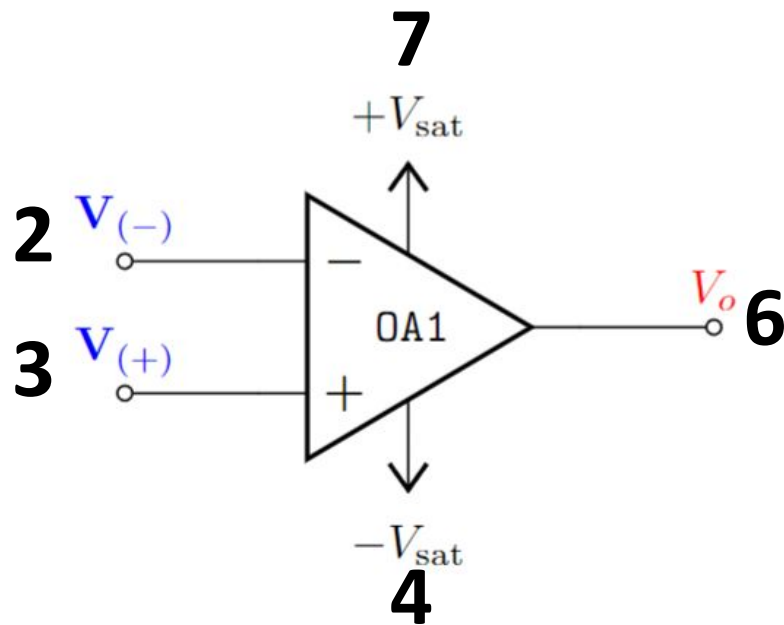
$$AV_d = (2 \times 10^5) \times (-0.5 \times 10^{-3}) \text{ V} = -100 \text{ V}$$

Since $AV_d < -15 \text{ V}$ (Negative saturation)

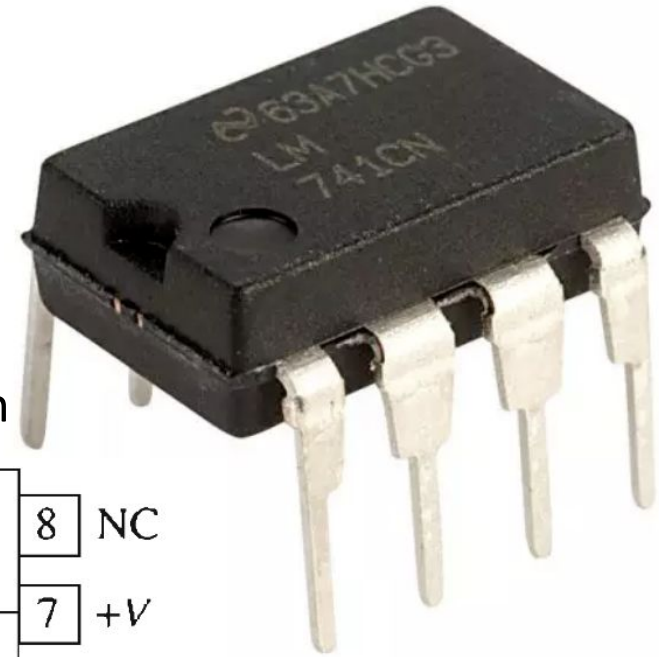
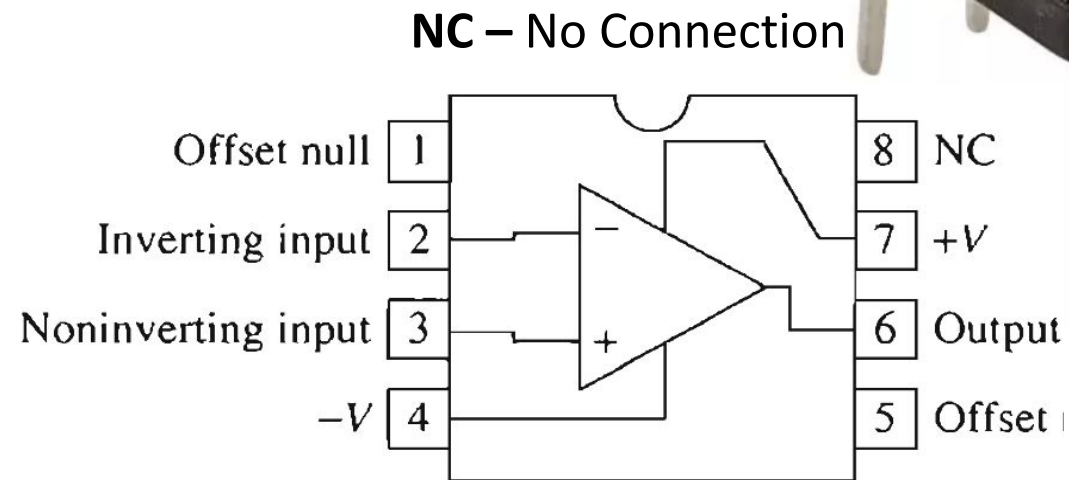
$$V_o = AV_d = -15 \text{ V}$$

Op-Amp: Physical Entity

Difference Amplifier – Amplifies the voltage difference between two terminals.



Circuit symbol for the general-purpose op amp.
Pin numbering is that for an **8-pin mini-DIP package**



μ A-741C

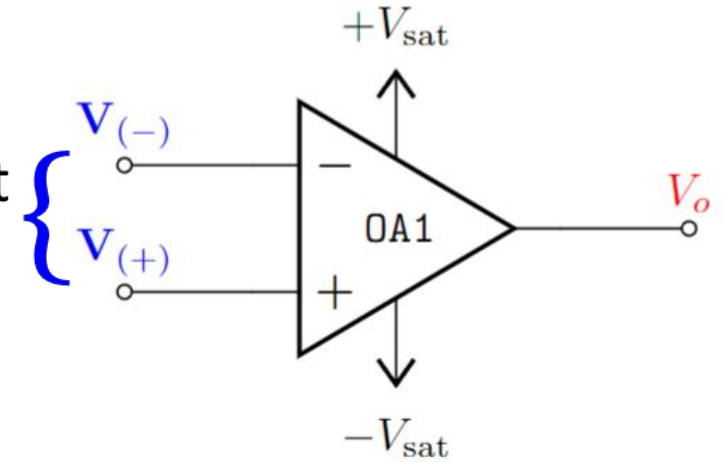
Manufacturer ID

Part Identification Number (PIN)

C: Commercial ($0^{\circ} - 70^{\circ}\text{C}$)
I: Industrial ($-25^{\circ} - 85^{\circ}\text{C}$)
M: Military ($-55^{\circ} - 125^{\circ}\text{C}$)

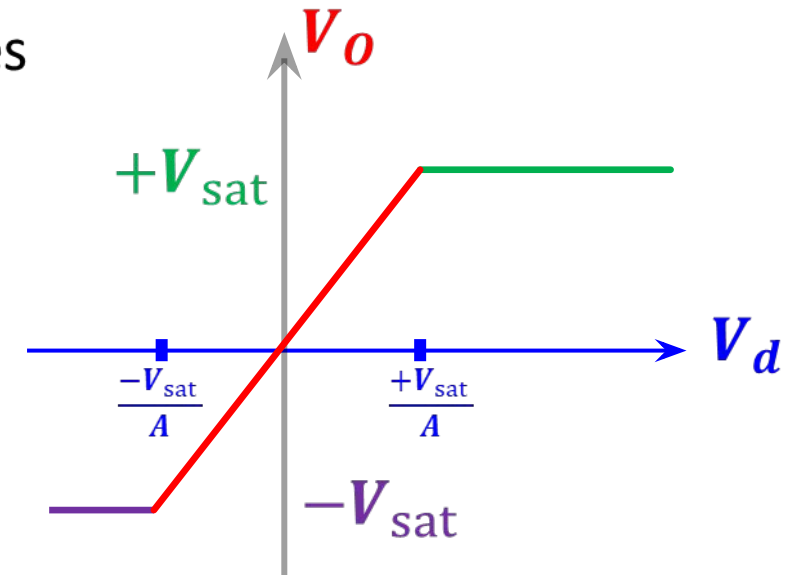
Op-Amp: Summary

Op-Amp **Amplifies** the difference between the voltages at its two input terminals - V_d

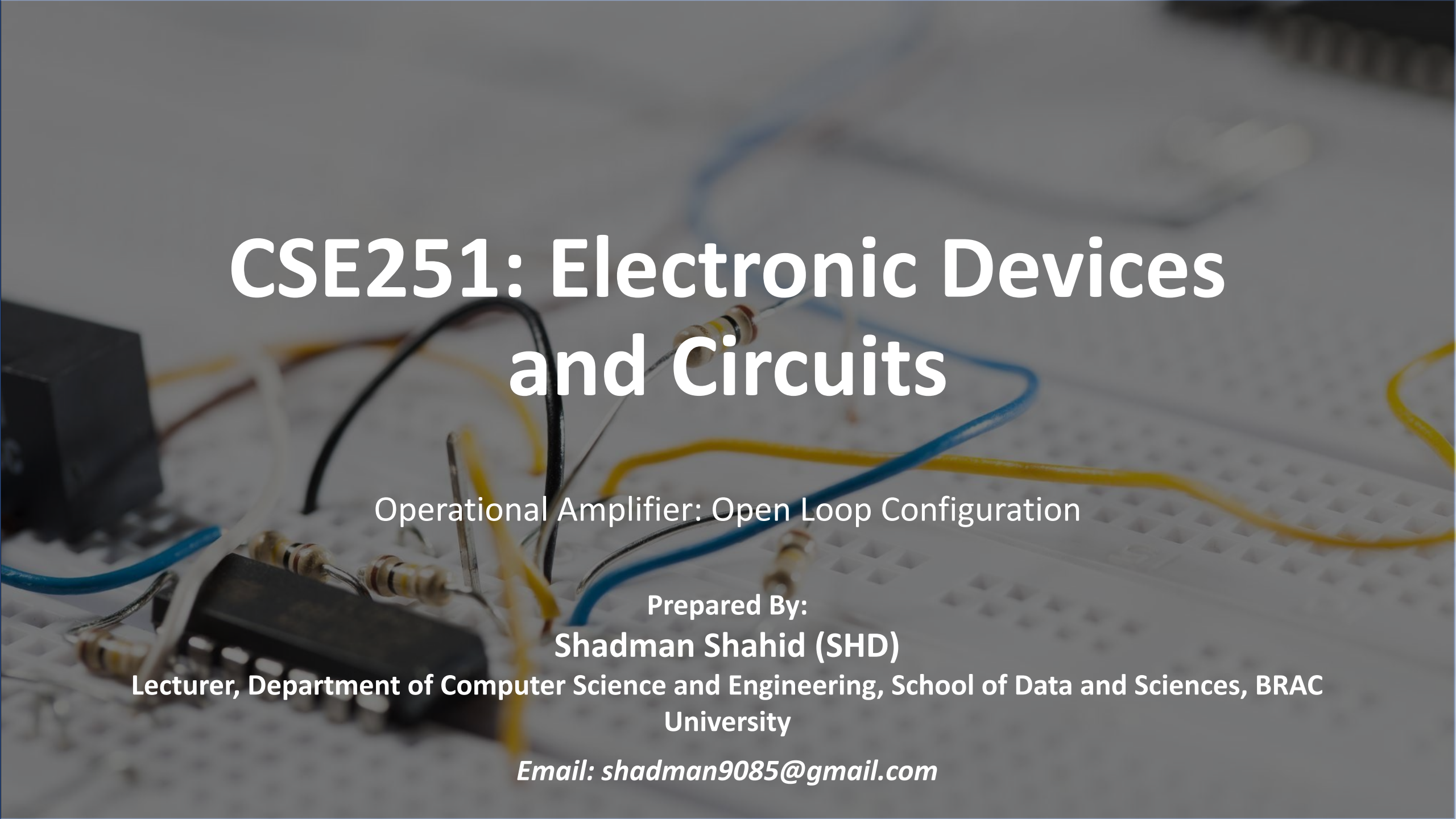


However, the **Amplification** is limited within voltage levels defined by the positive and negative saturation voltages $[-V_{sat}, +V_{sat}]$.

The “ideal” op-amp behaves like a **voltage dependent voltage source** within the linear region.



Thank you!



CSE251: Electronic Devices and Circuits

Operational Amplifier: Open Loop Configuration

Prepared By:

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Outline

- Operational Amplifier – Terminals and VTC (Recap)
- Op-Amp: Circuit Modelling
- Example Problem – Op-amp model

- Op-amp configuration – open and closed loop
- Open Loop op-amp: Voltage Transfer Characteristics
- Open Loop op-amp: Comparators
- Comparator Application
 - Smoke detector
 - Automatic AC switch

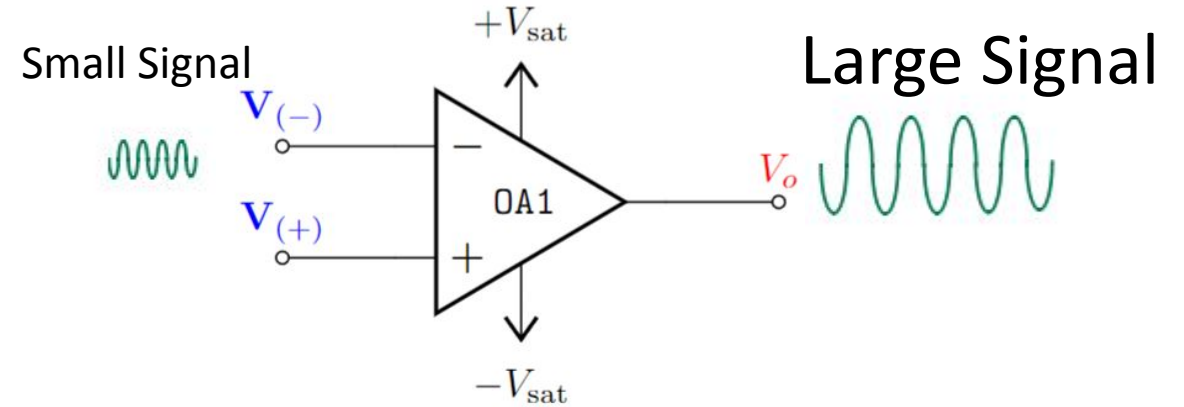
Op-Amp (Recap)

- **Operational:**

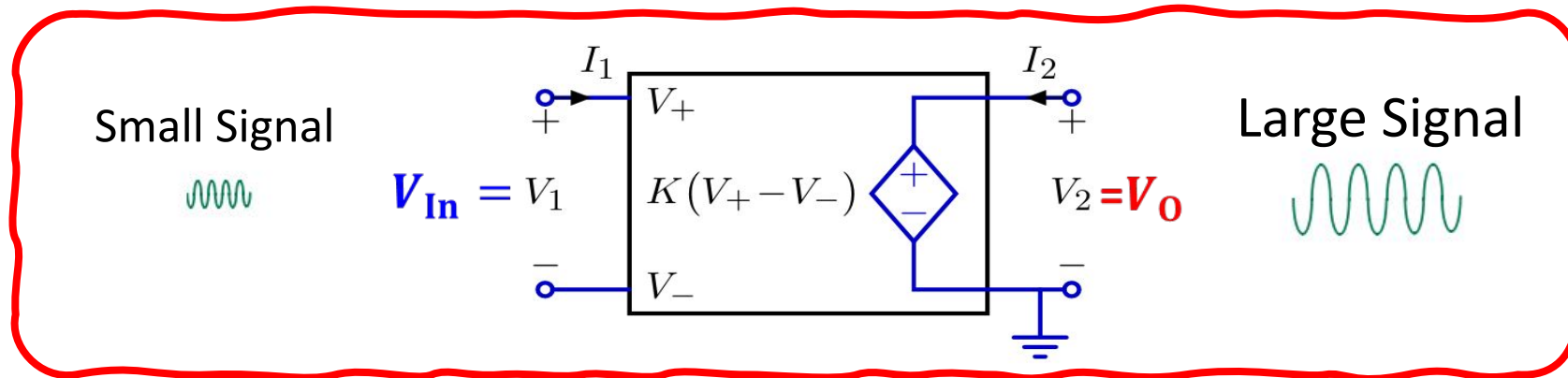
Mathematical Operations

- **Amplifier:**

Amplifies input signal/voltage.



An **op-amp (operational amplifier)** can be represented by a **voltage- controlled voltage source**.



Op-Amp: VTC (Recap)

Voltage Transfer Characteristics (VTC)

Positive saturation:

If $V_d > \frac{+V_{\text{sat}}}{K}$: Positive

$$\Rightarrow V_o = +V_{\text{sat}}$$

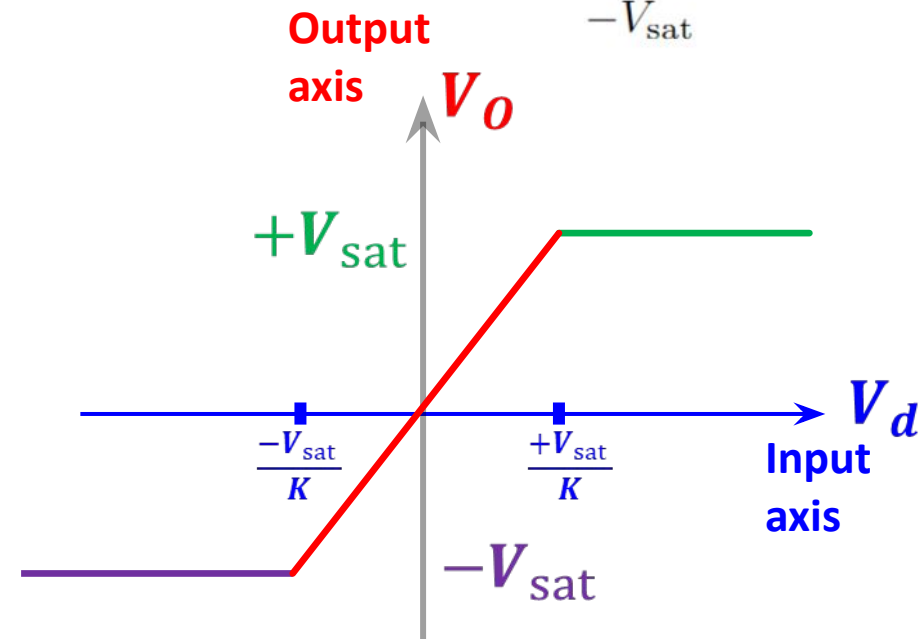
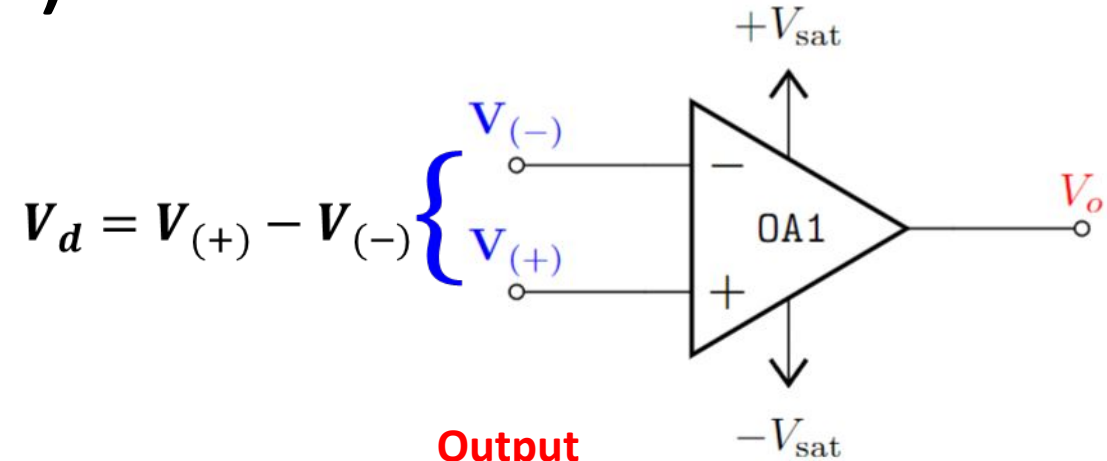
Linear Region

$V_o = KV_d$: When V_d is very small
 $-V_{\text{sat}} < V_o = KV_d < +V_{\text{sat}}$

Negative saturation:

If $V_d < \frac{-V_{\text{sat}}}{K}$: Negative

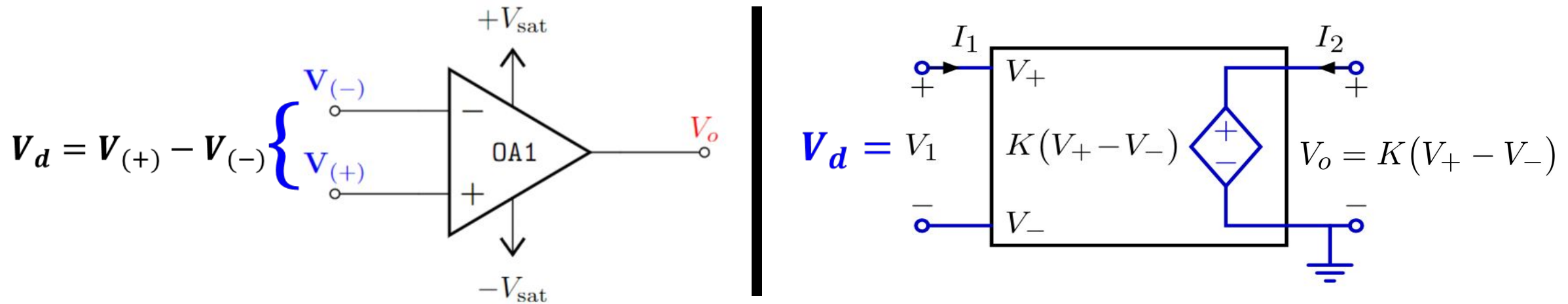
$$\Rightarrow V_o = -V_{\text{sat}}$$



Op-Amp: Circuit Modelling

Voltage controlled voltage Source

“Ideal” op-amp approximation



$$V_{(+)} \approx V_{(-)}$$

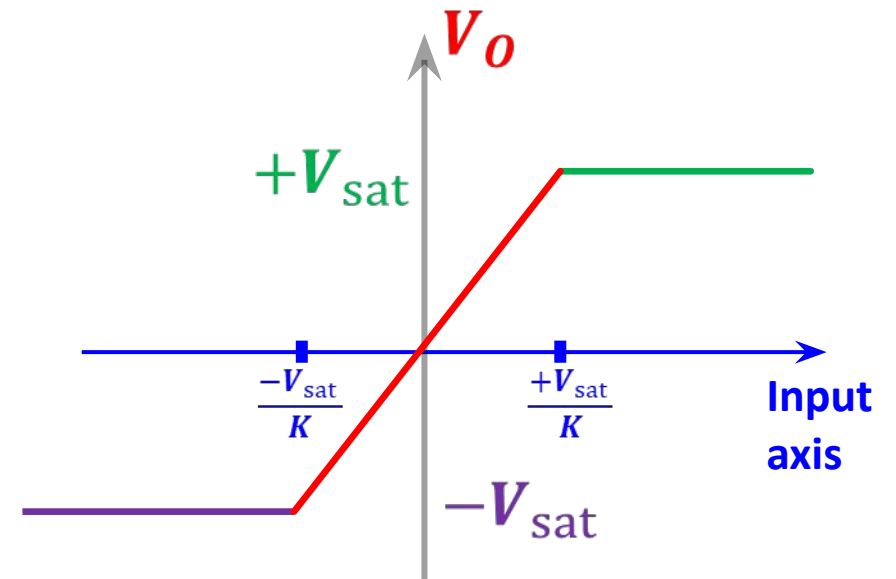
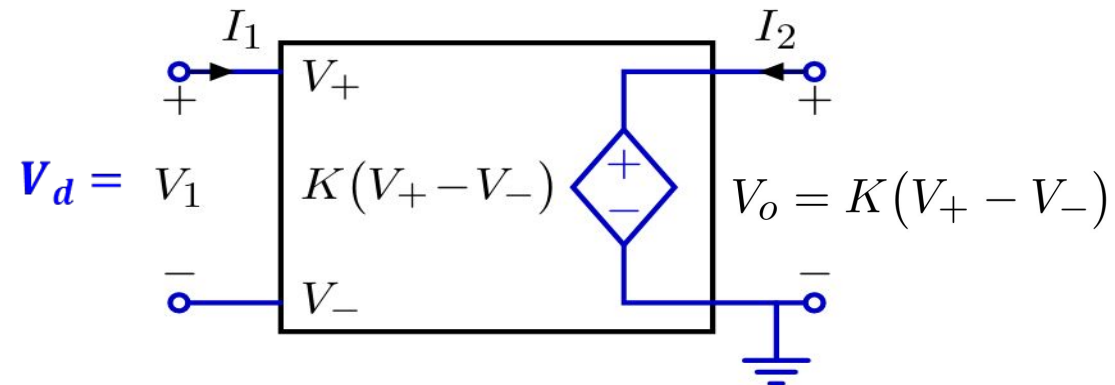
$$I_1 \approx 0$$

Op-amp: Circuit Model and VTC

- **Voltage (Differential/OL) Gain:** A/A_{OL} or K Slope of **VTC**

Parameter	Typical Range	Ideally

- Positive Saturation Voltage: $+V/V_{CC}/V_{sat}$
- Negative Saturation Voltage: $-V/V_{EE}/-V_{sat}$



Op-amp: Circuit Model and VTC

Voltage Transfer Characteristics (VTC)

Positive saturation:

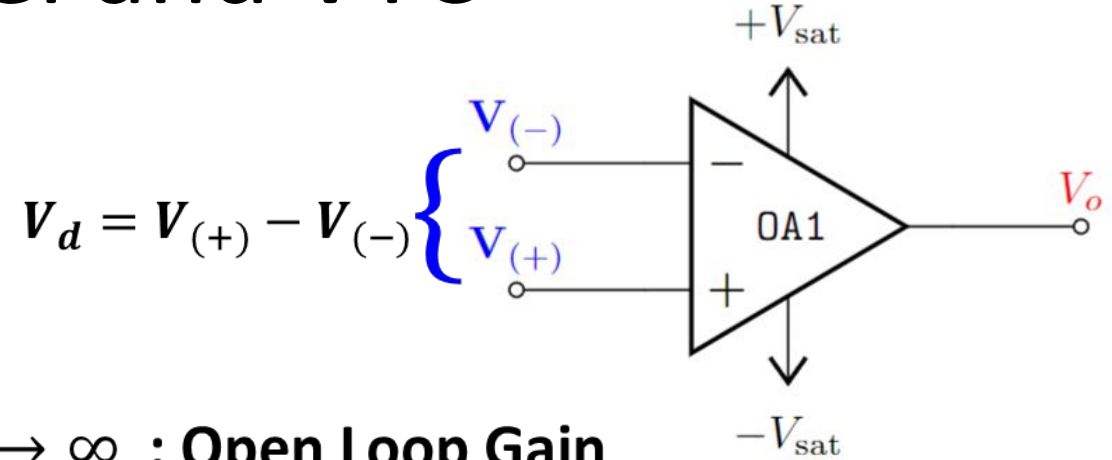
$$\Rightarrow V_O = +V_{\text{sat}}$$

Linear Region

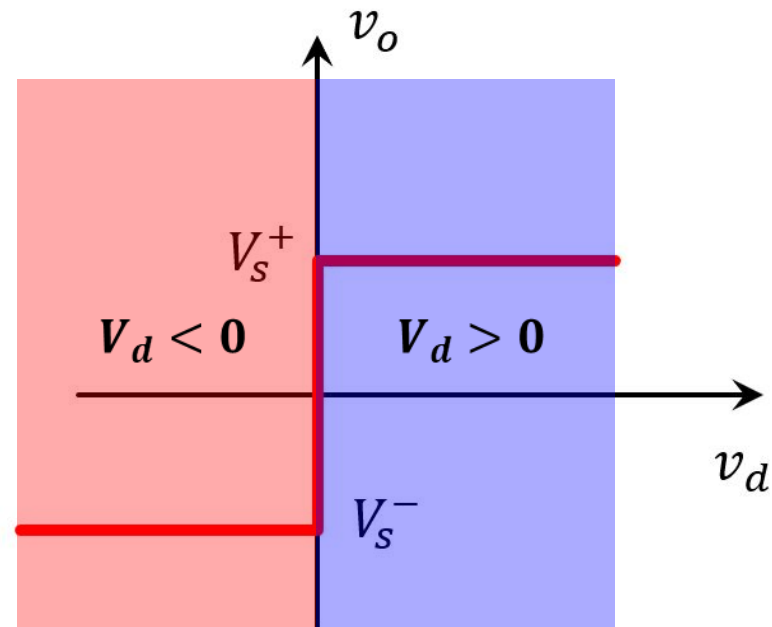
$$-V_{\text{sat}} < V_O = AV_d < +V_{\text{sat}}$$

Negative saturation:

$$\Rightarrow V_O = -V_{\text{sat}}$$

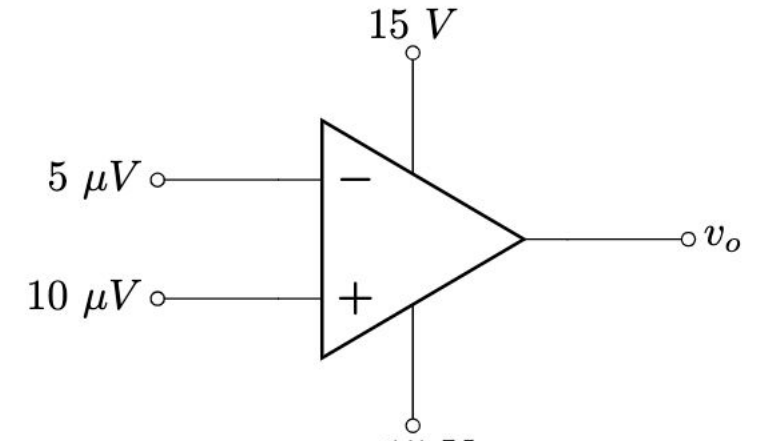


$A \rightarrow \infty$: Open Loop Gain

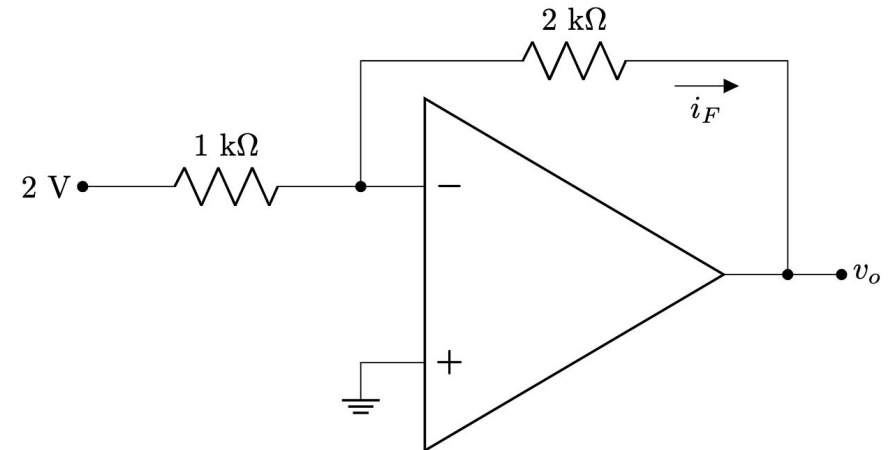


Types of Op-Amp configuration

1. Open loop configuration:
No physical connection between input and output

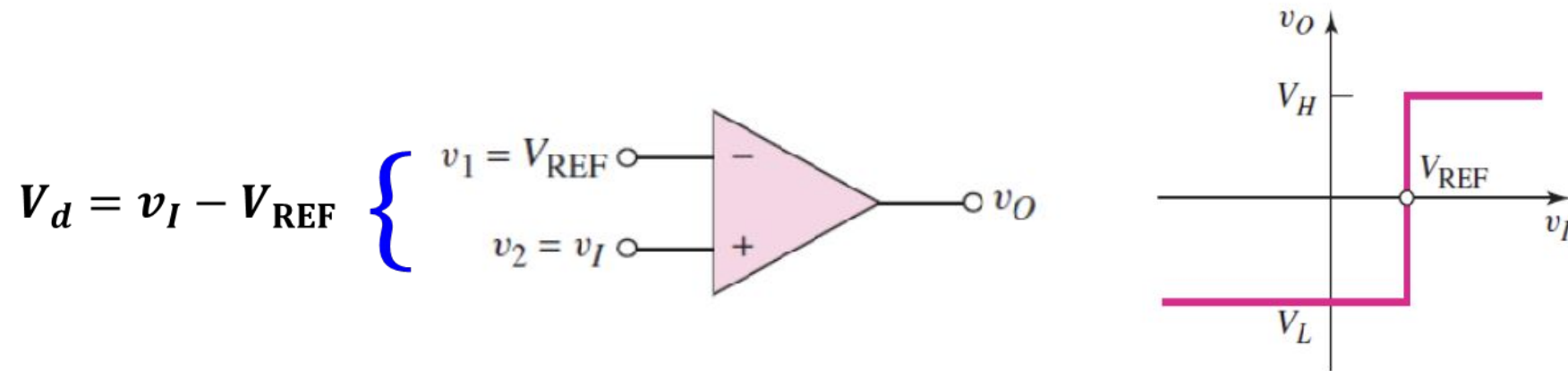


1. Closed loop configuration:
Feedback from output terminal



Open Loop Configuration: Comparator

Level Crossing Detector / Comparator



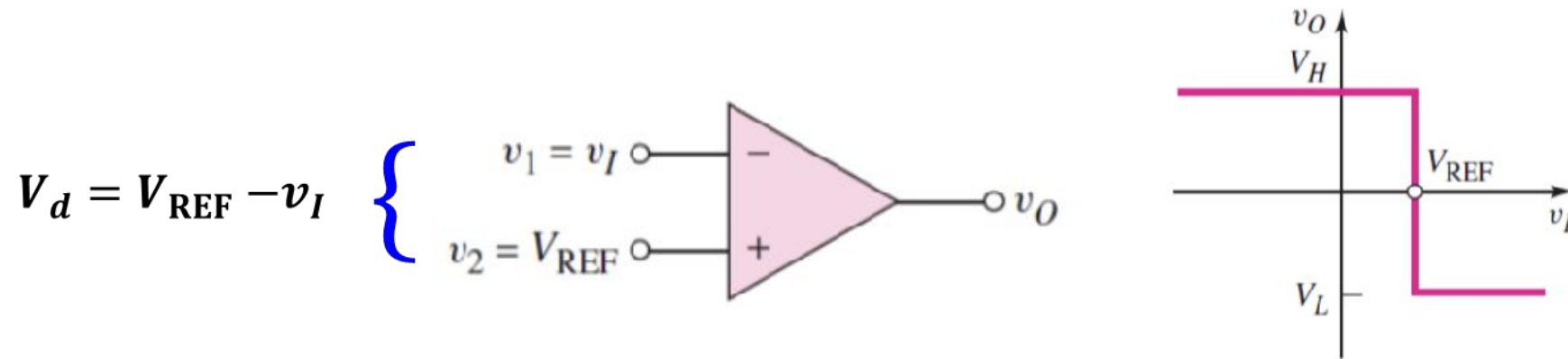
NON-INVERTING COMPARATOR

$$V_d = v_I - V_{\text{REF}} > 0 \quad \Rightarrow \quad v_O = V_H$$

$$v_I > V_{\text{REF}} \quad \Rightarrow \quad v_O = V_H$$

Open Loop Configuration: Comparator

Level Crossing Detector / Comparator



INVERTING COMPARATOR

$$V_d = V_{\text{REF}} - v_I > 0 \quad \Rightarrow \quad v_O = V_H$$

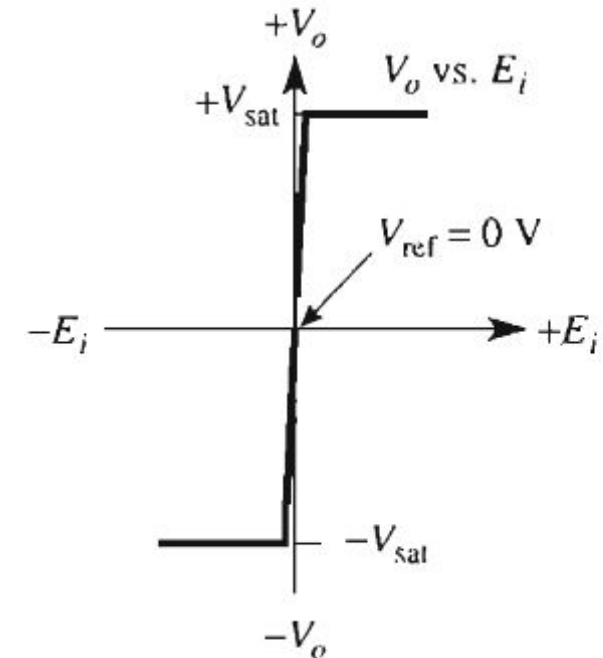
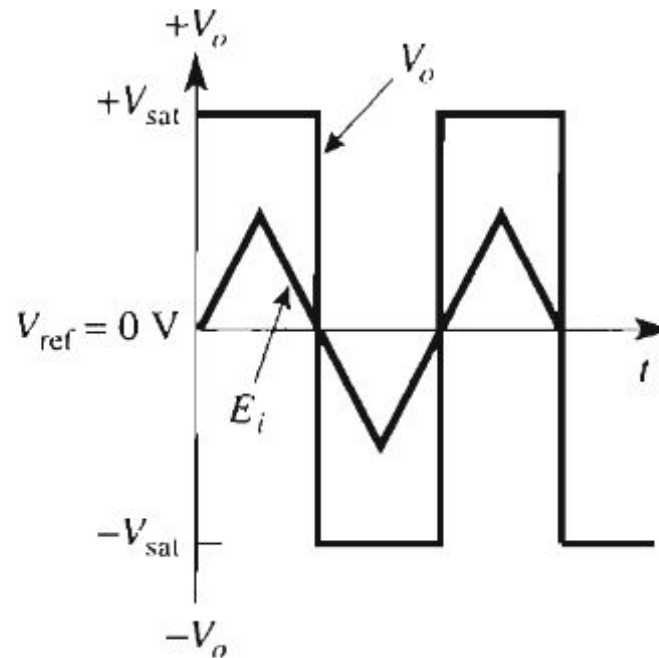
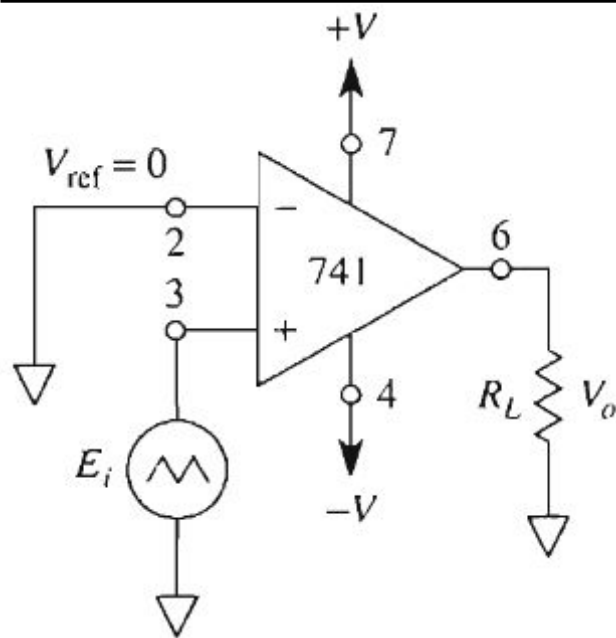
$$v_I < V_{\text{REF}} \quad \Rightarrow \quad v_O = V_H$$

Open Loop Configuration: Comparator

Zero Crossing Detector

Compare values with a reference and pin value to $+V_{sat}$ if voltage is above or to below that.

Non-inverting configuration

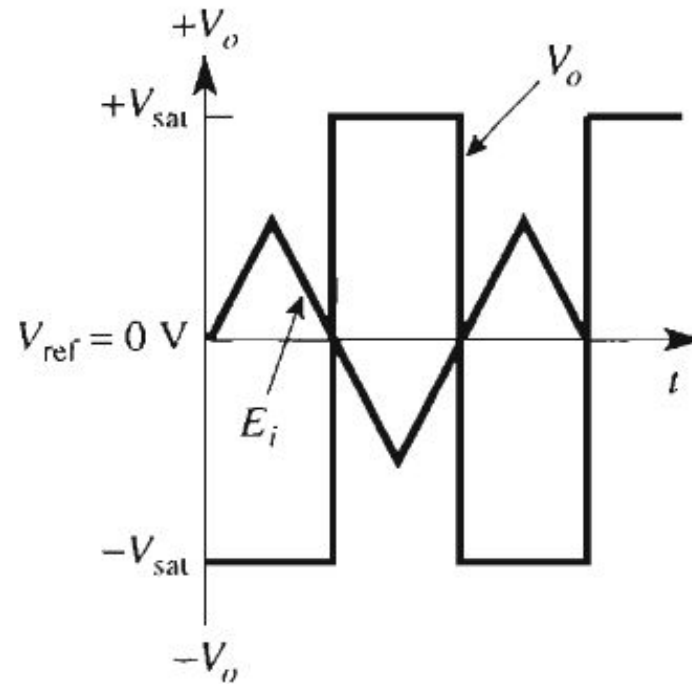
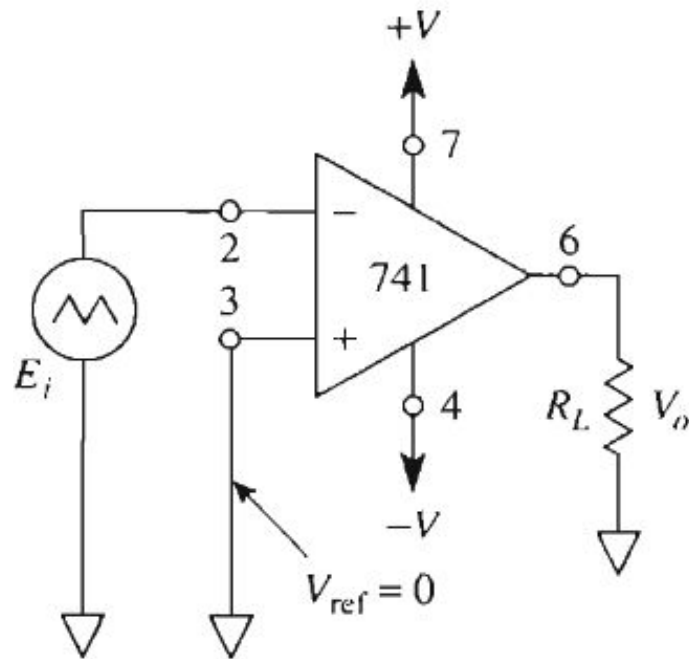


(a) Noninverting: When E_i is above V_{ref} , $V_o = +V_{sat}$.

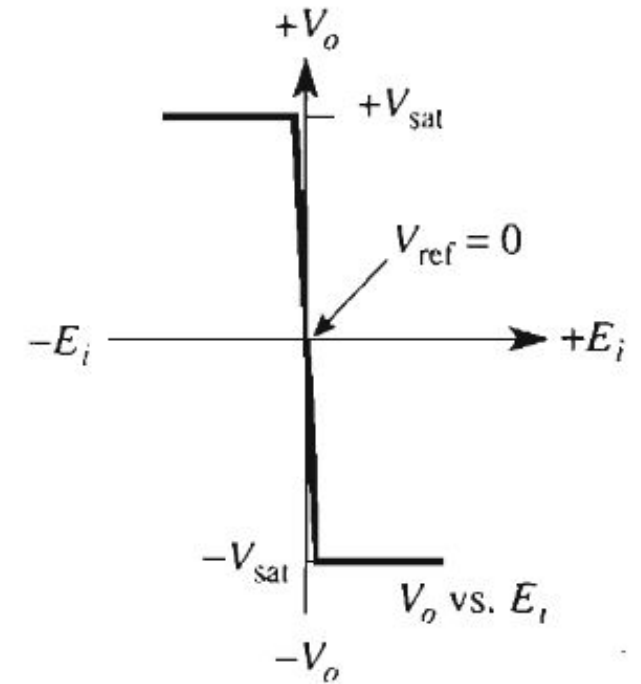
Open Loop Configuration: Comparator

Zero Crossing Detector

Inverting configuration



(b) Inverting: When E_i is above V_{ref} , $V_o = -V_{sat}$.



Summary

Level Crossing Detector / Comparator

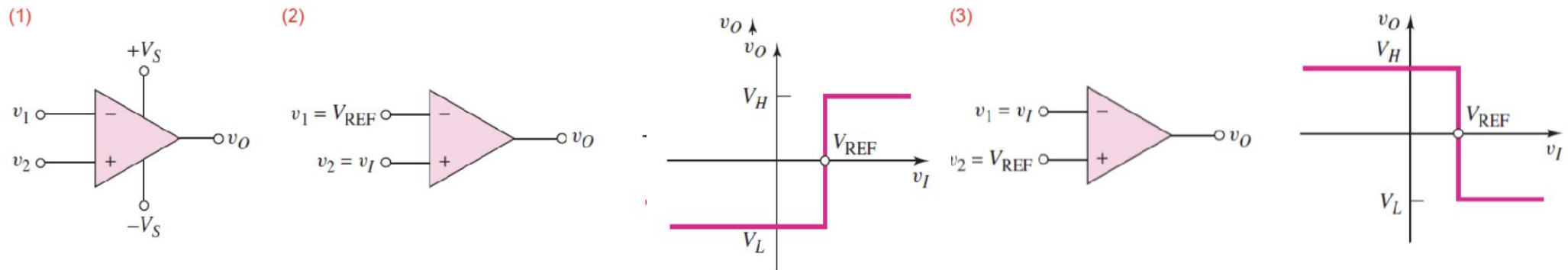
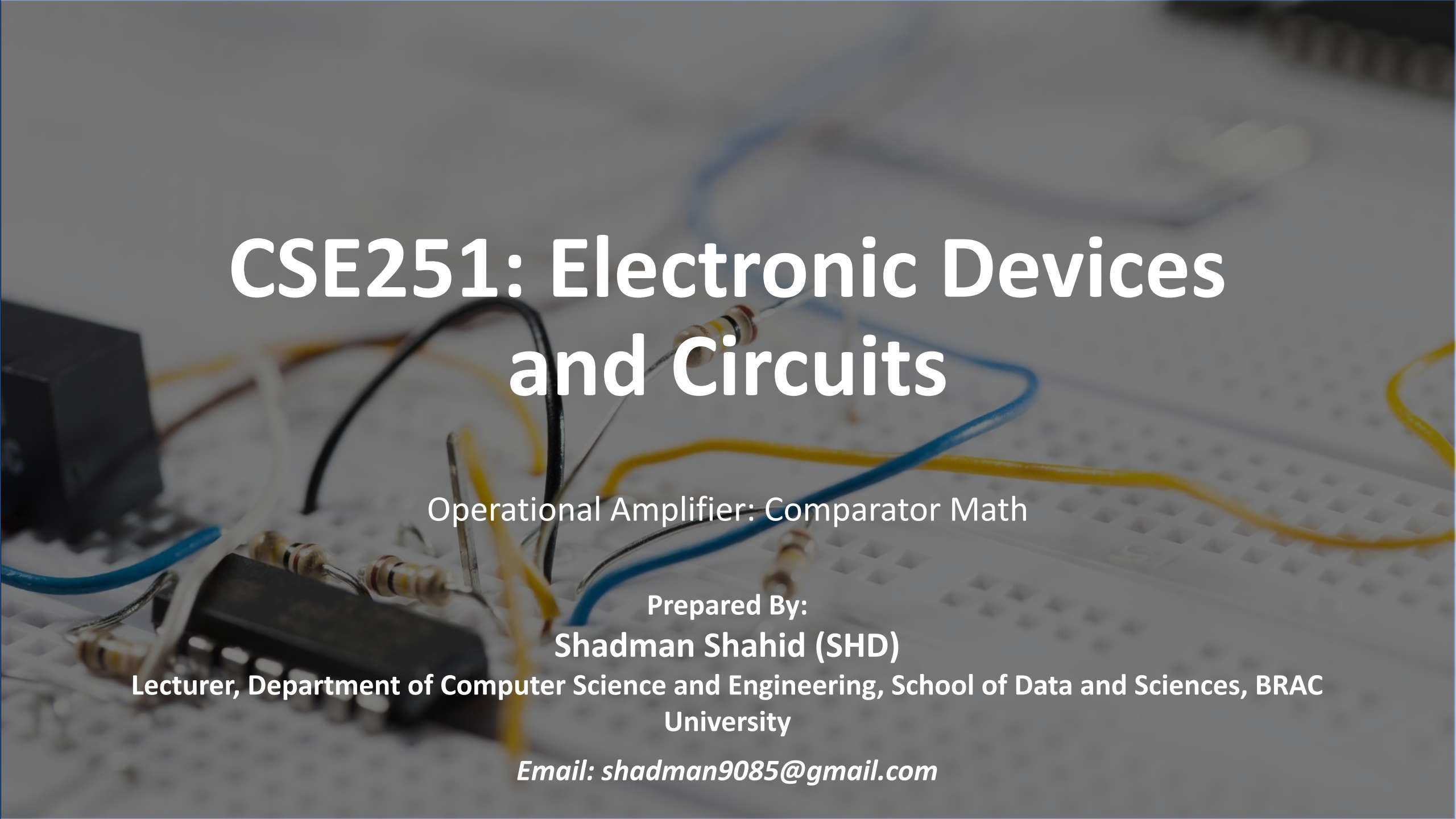


Figure 2: (1) Op-Amp Comparator (2) Noninverting Circuit (3) Inverting Circuit

Thank you



CSE251: Electronic Devices and Circuits

Operational Amplifier: Comparator Math

Prepared By:

Shadman Shahid (SHD)

Lecturer, Department of Computer Science and Engineering, School of Data and Sciences, BRAC
University

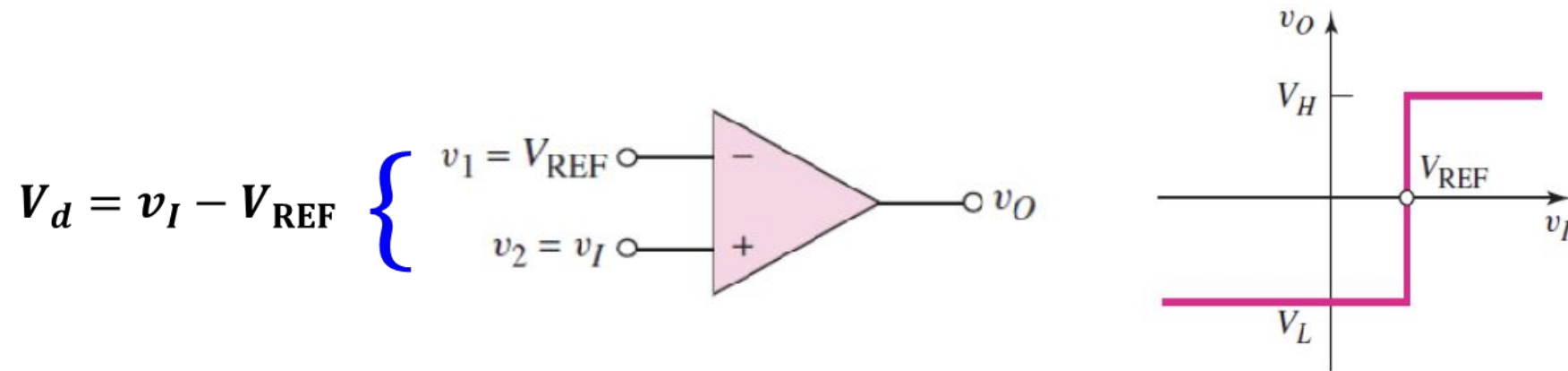
Email: shadman9085@gmail.com

Outline

- Op-Amp: Comparator (Recap)
- Open Loop Configuration: Example 1
- Open Loop Configuration: Example 2
- Open Loop Configuration: Example 3

Op-Amp: Comparator (Recap)

Level Crossing Detector / Comparator



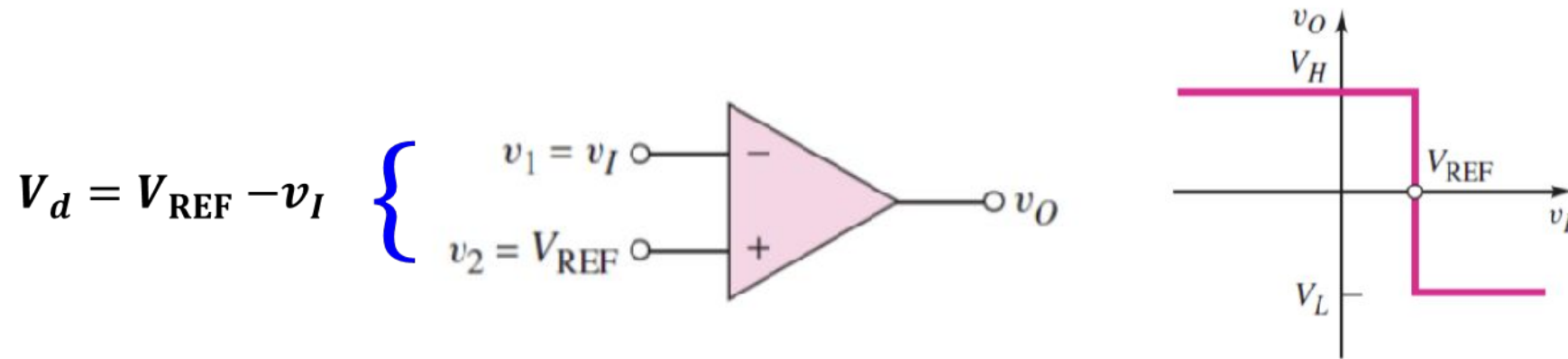
NON-INVERTING COMPARATOR

$$V_d = v_I - V_{\text{REF}} > 0 \quad \Rightarrow \quad v_O = V_H$$

$$v_I > V_{\text{REF}} \quad \Rightarrow \quad v_O = V_H$$

Op-Amp: Comparator (Recap)

Level Crossing Detector / Comparator



INVERTING COMPARATOR

$$V_d = V_{\text{REF}} - v_I > 0 \quad \Rightarrow \quad v_O = V_H$$

$$v_I < V_{\text{REF}} \quad \Rightarrow \quad v_O = V_H$$

Op-Amp: Comparator (Recap)

Level Crossing Detector / Comparator

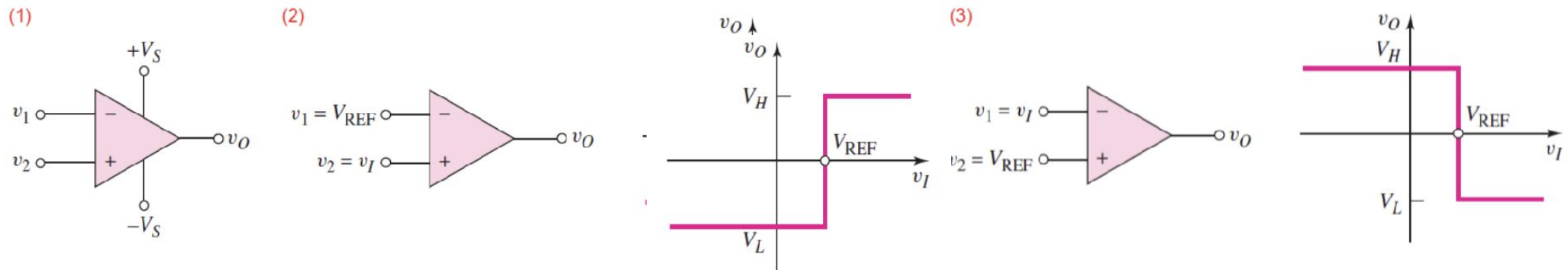
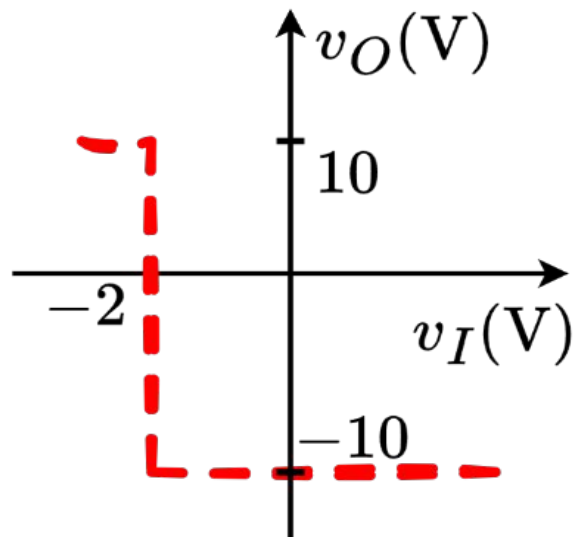


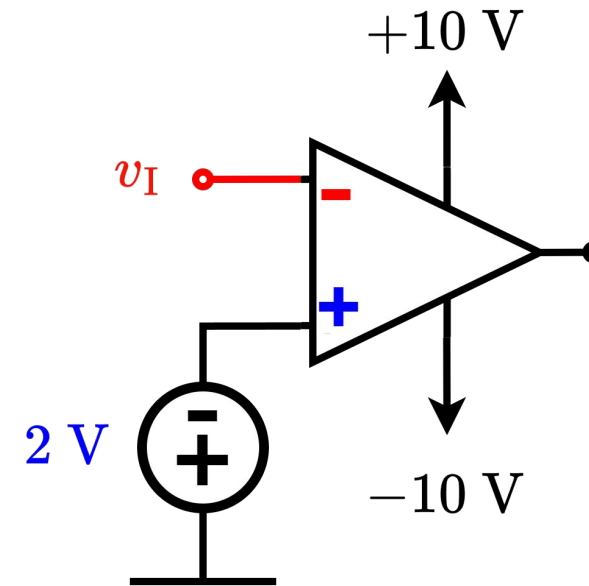
Figure 2: (1) Op-Amp Comparator (2) Noninverting Circuit (3) Inverting Circuit

Open Loop Configuration: Example 1

- Design a circuit using **op-amp** that has the voltage transfer characteristics as shown in the figure below. $v_o(V)$ is the **output voltage** and $v_i(V)$ is the **input voltage**.



Solution:



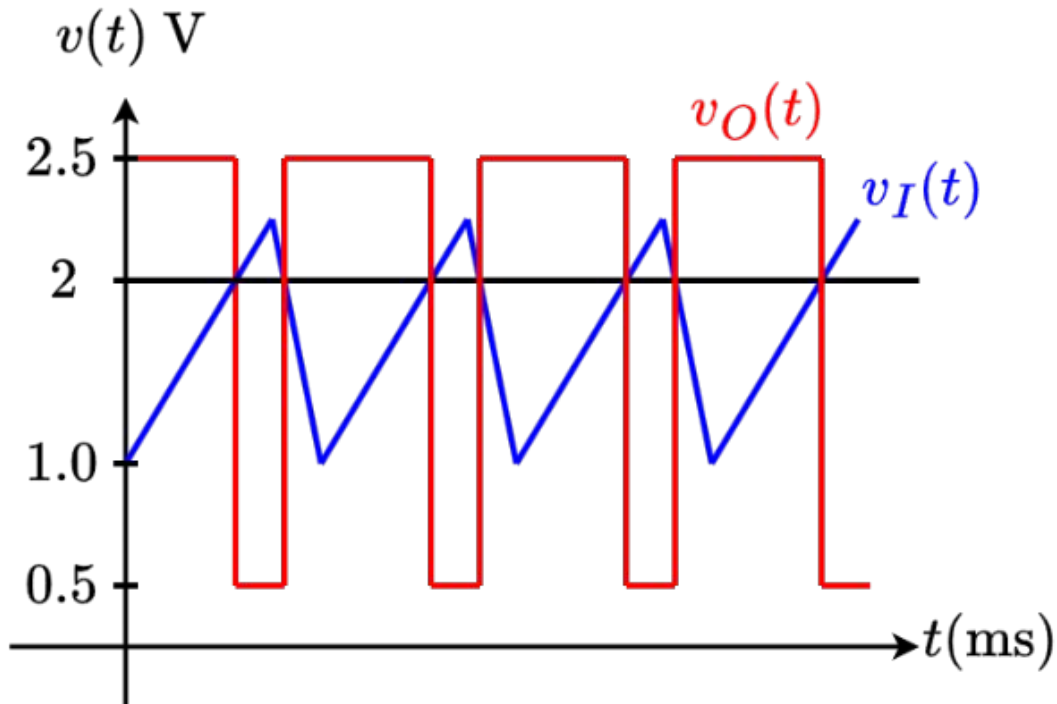
Open Loop Configuration: Example 2

- Draw the voltage transfer characteristic (VTC) curve (v_O vs v_I) from the adjacent waveform graph. Also draw the **Op-Amp Circuit** that would give rise to such a VTC.

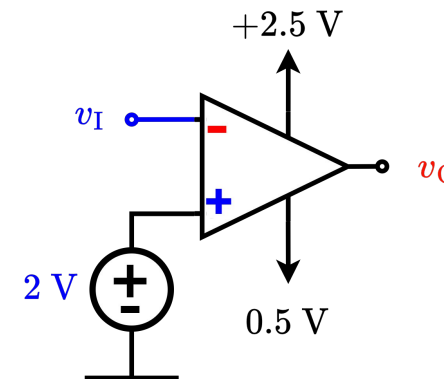
Solution:

v_I smaller than 2 V $\Rightarrow v_O = 2.5$ V $\rightarrow$ **Positive Saturation**

v_I larger than 2 V $\Rightarrow v_O = -2.5$ V $\rightarrow$ **Negative Saturation**

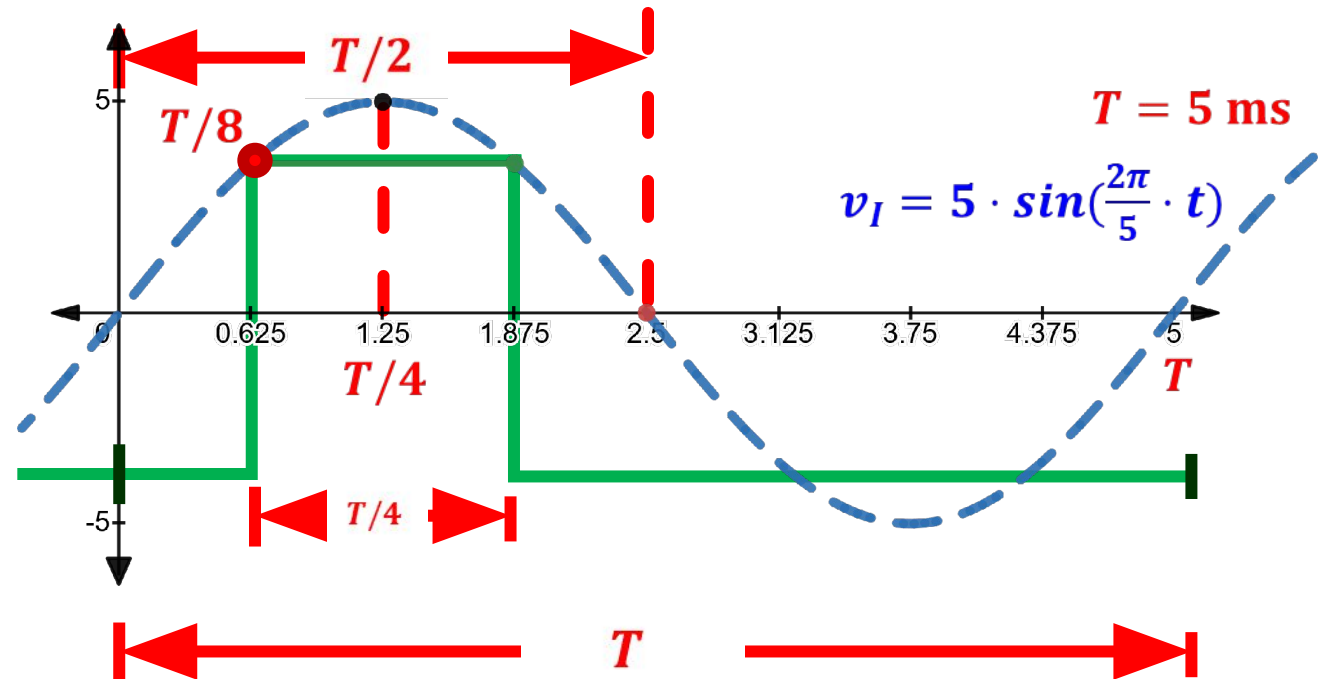
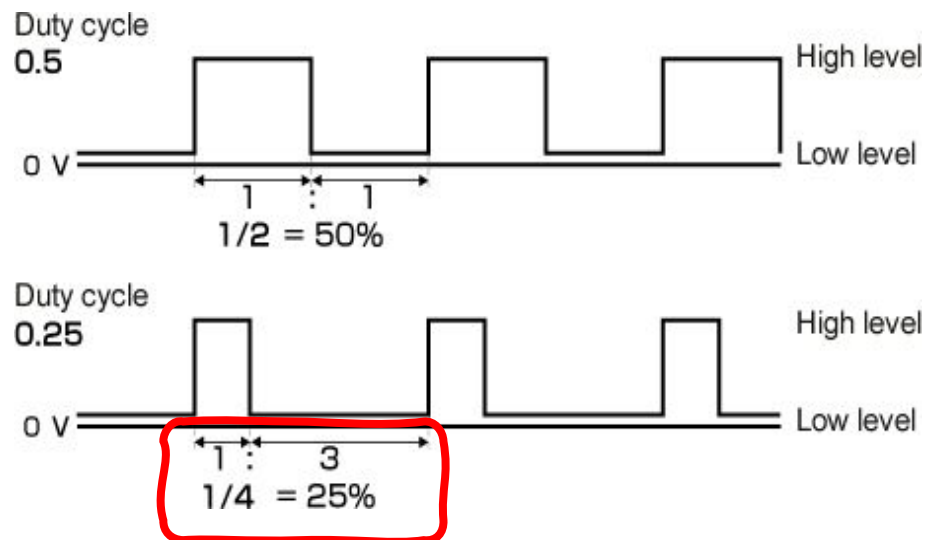


INVERTING CONFIGURATION



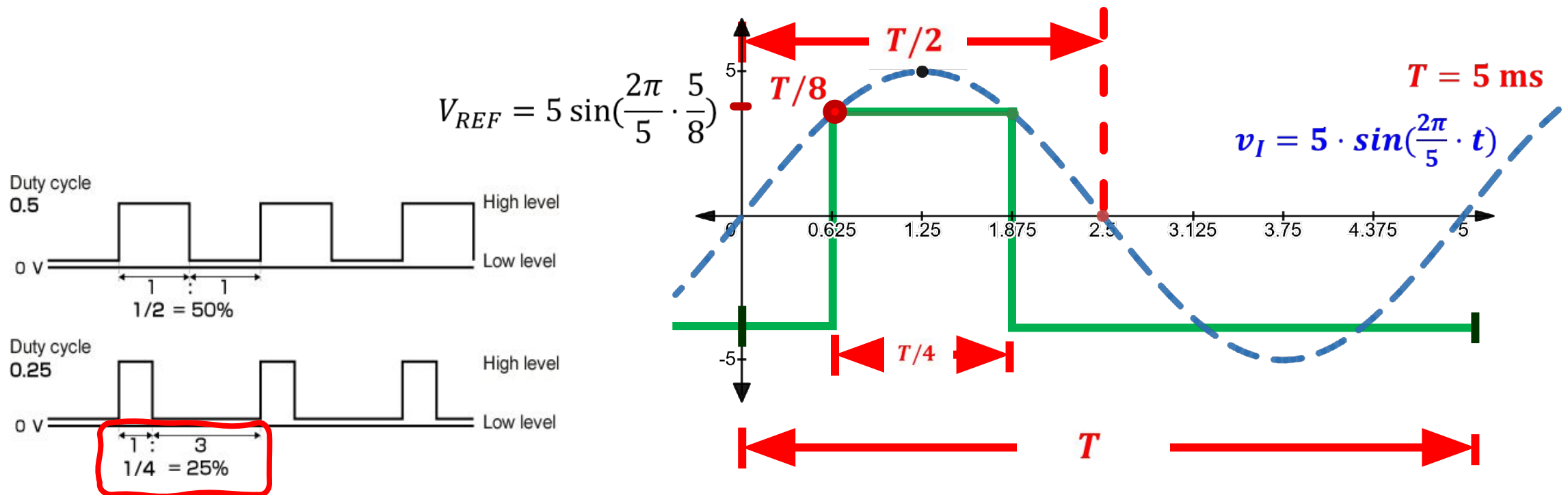
Open Loop Configuration: Example 3

Design an op-amp circuit to transform the sinusoidal voltage, $v_I = 5 \cdot \sin\left(\frac{2\pi}{5} \cdot t\right)$ (t is in units of ms , and time-period T is $5\ ms$), to: A square wave with a duty cycle of **25%**.



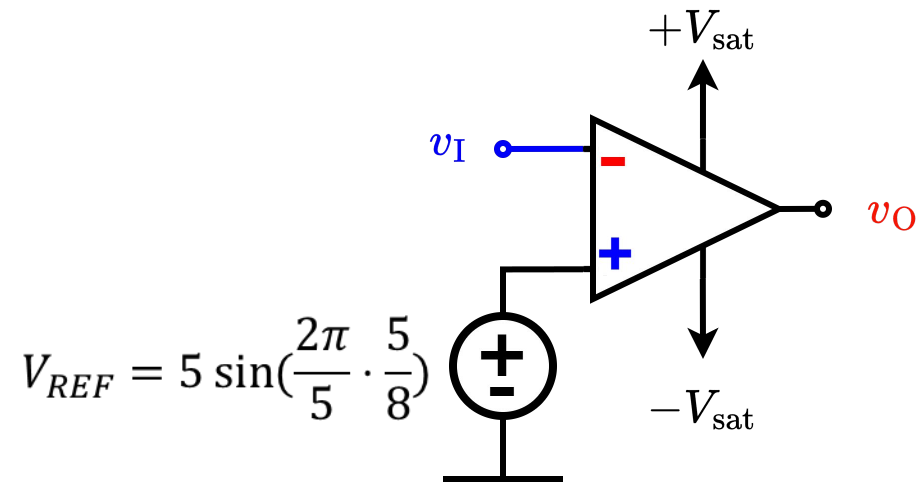
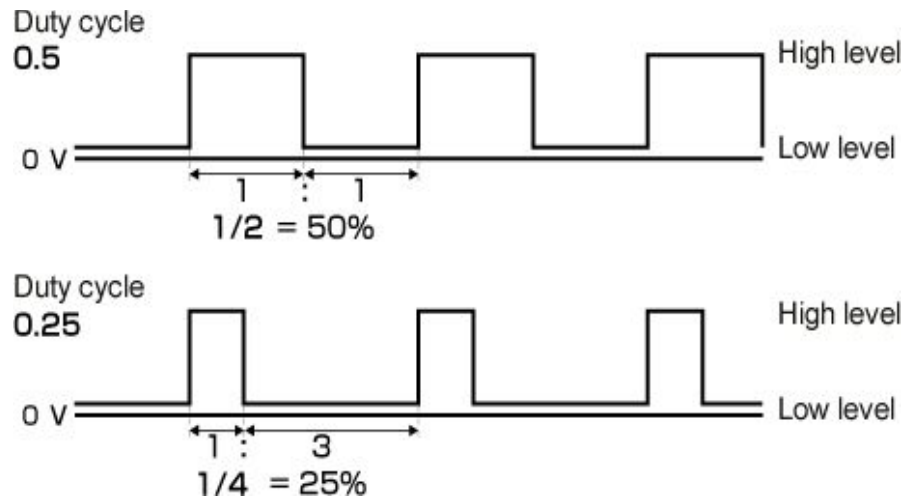
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Open Loop Configuration: Example 3

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Thank you



CSE251: Electronic Devices and Circuits

Operational Amplifier: Comparator Application

Prepared By:

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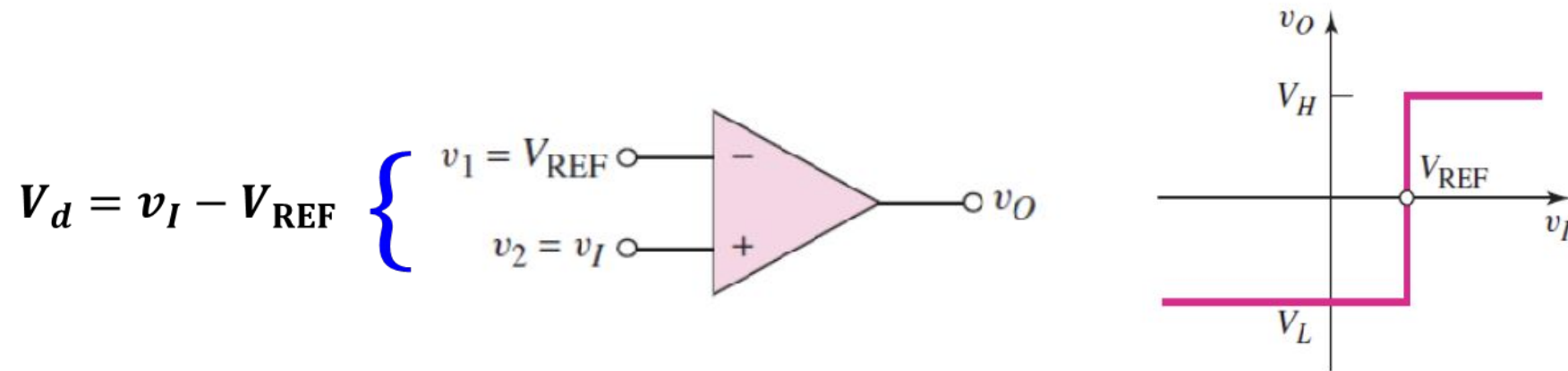
Email: shadman9085@gmail.com

Outline

- Op-Amp: Comparator (Recap)
- Comparator Application - Smoke Detectors
- Smoke Detector – Operation
- Comparator Application – Automatic AC
- General Principle - Comparator

Op-Amp: Comparator (Recap)

Level Crossing Detector / Comparator



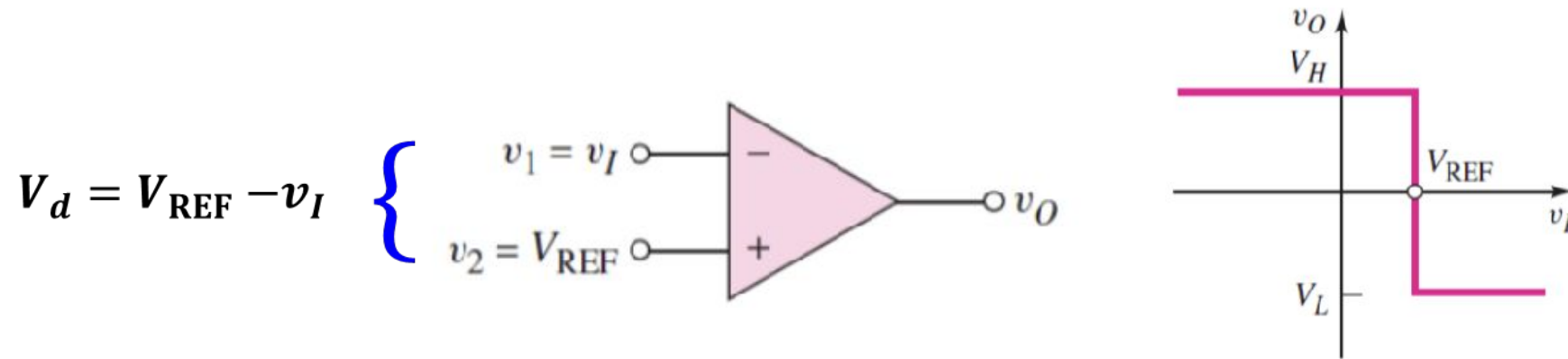
NON-INVERTING COMPARATOR

$$V_d = v_I - V_{\text{REF}} > 0 \quad \Rightarrow \quad v_O = V_H$$

$$v_I > V_{\text{REF}} \quad \Rightarrow \quad v_O = V_H$$

Op-Amp: Comparator (Recap)

Level Crossing Detector / Comparator



INVERTING COMPARATOR

$$V_d = V_{\text{REF}} - v_I > 0 \quad \Rightarrow \quad v_O = V_H$$

$$v_I < V_{\text{REF}} \quad \Rightarrow \quad v_O = V_H$$

Comparator Application - Smoke Detectors

Working Principle:

Smoke present → Alarm rings

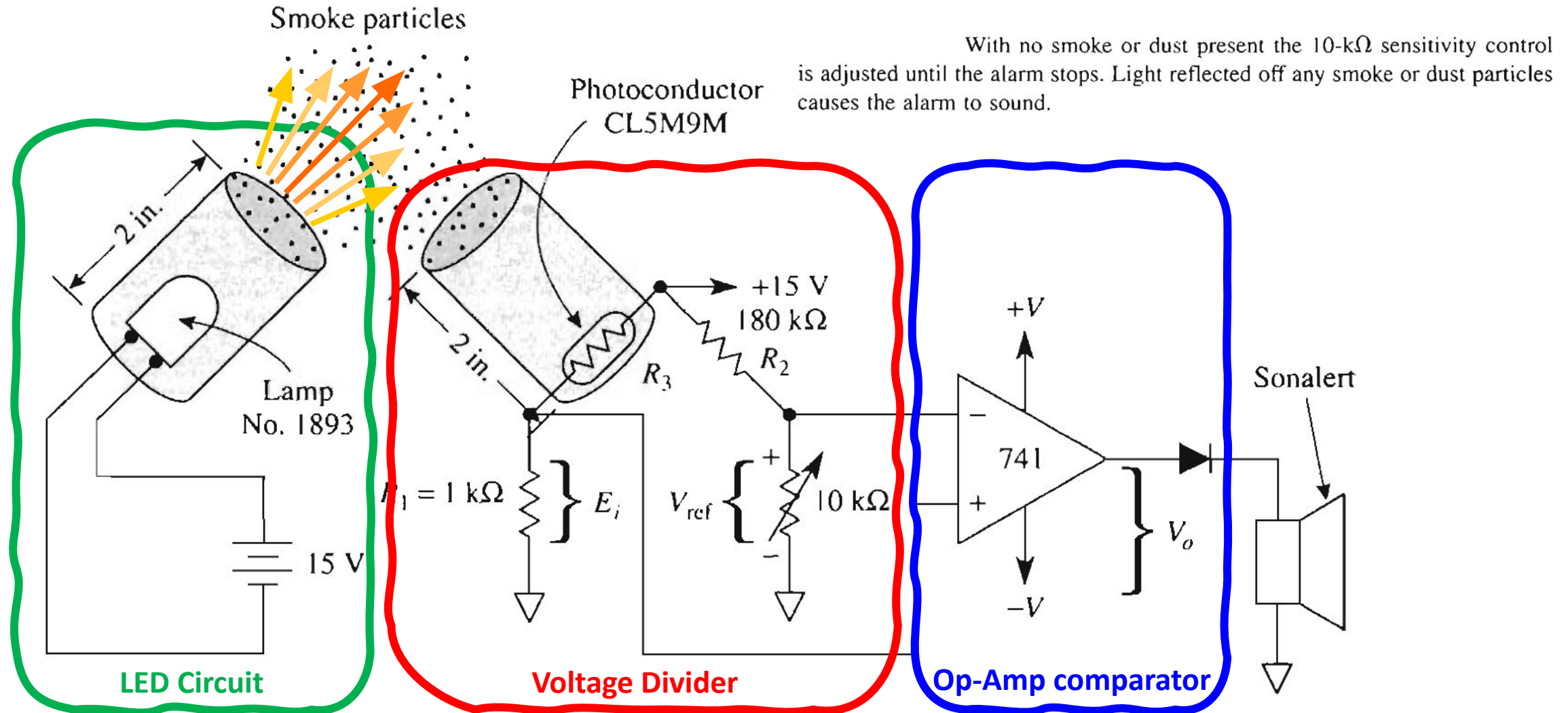
No smoke present → No Alarm rings.



Since the **alarm** is an electrically driven device, we need some method to convert smoke levels to proportionate voltage.

Smoke levels → Voltage

Comparator Application - Smoke Detectors



Smoke Detector - Operation

Light from the LED is reflected off the smoke particles on to the photoconductor –

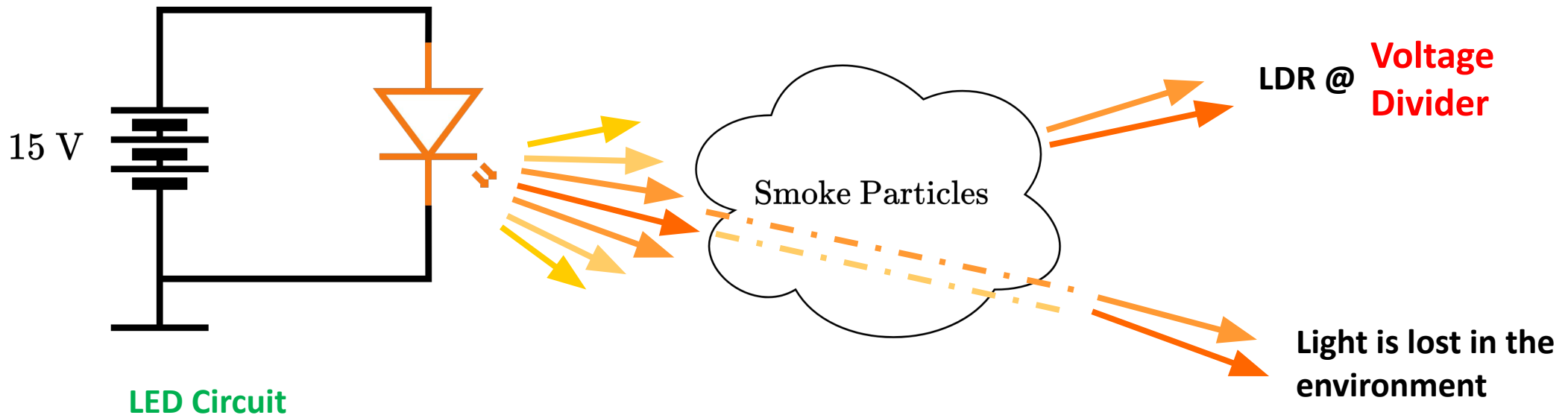
Light Dependent Resistor (LDR).

• **No Smoke**

- No scattering
- Low intensity at LDR (Dark - $I \downarrow$)

• **Smoke Present**

- High scattering from smoke particles
- High intensity at LDR (Light - $I \uparrow$)



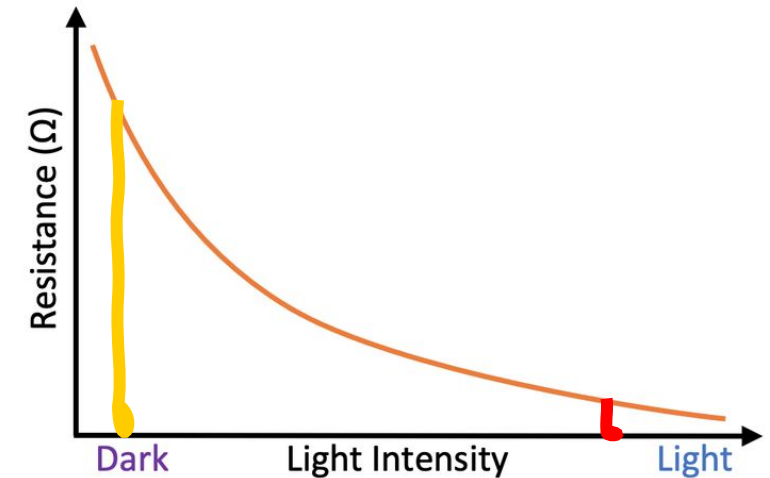
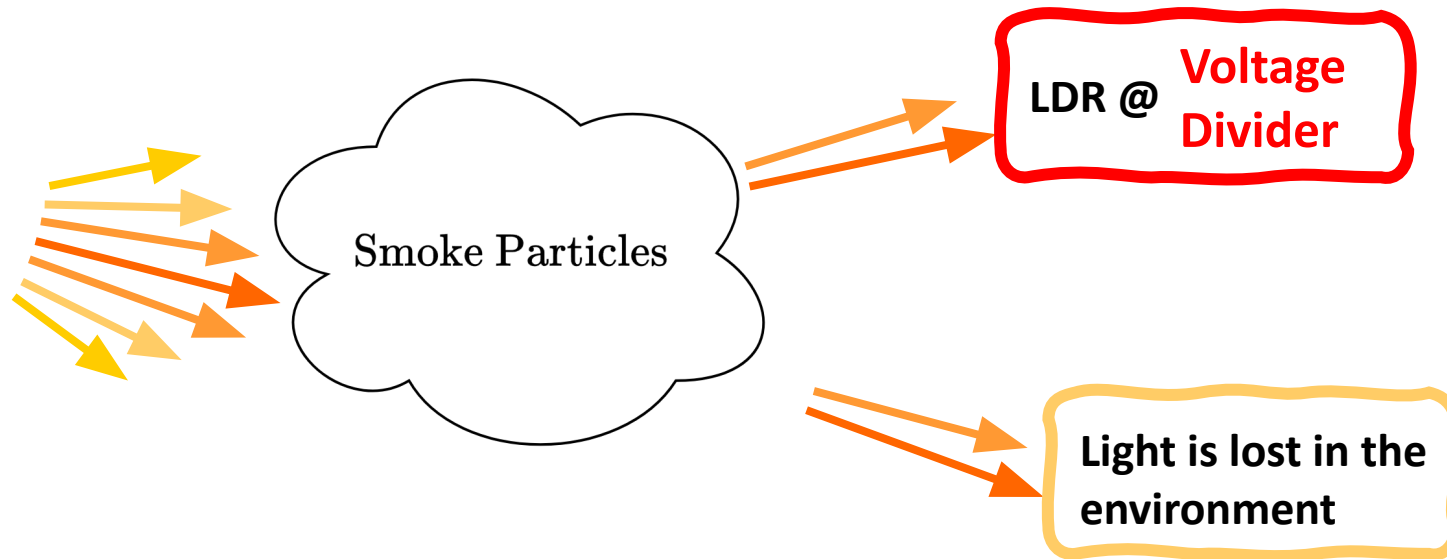
Smoke Detector - Operation

No Smoke

- No scattering
- Low intensity at LDR (Dark - $I \downarrow$)
- High resistance across LDR ($R_{LDR} \uparrow$)

Smoke Present

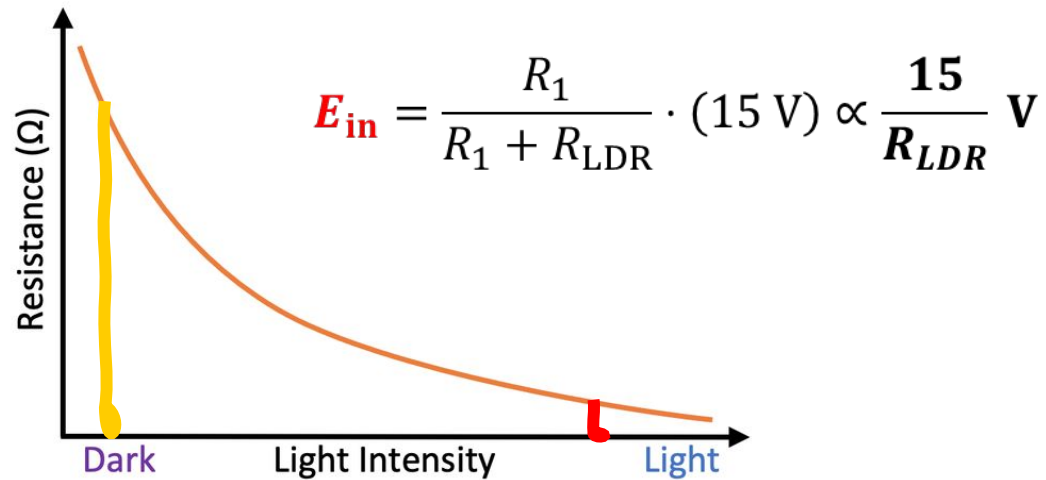
- High scattering from smoke particles
- High intensity at LDR (Light - $I \uparrow$)
- Low resistance across LDR ($R_{LDR} \downarrow$)



Smoke Detector - Operation

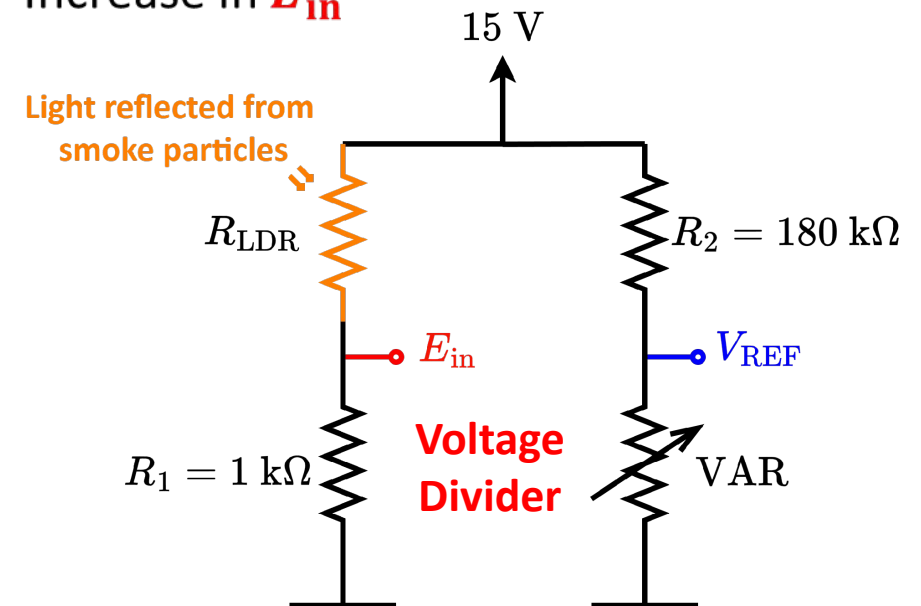
No Smoke

- No scattering
- Low intensity at LDR (Dark - $I \downarrow$)
- High resistance across LDR ($R_{LDR} \uparrow$)
- Decrease in E_{in}



Smoke Present

- High scattering from smoke particles
- High intensity at LDR (Light - $I \uparrow$)
- Low resistance across LDR ($R_{LDR} \downarrow$)
- Increase in E_{in}



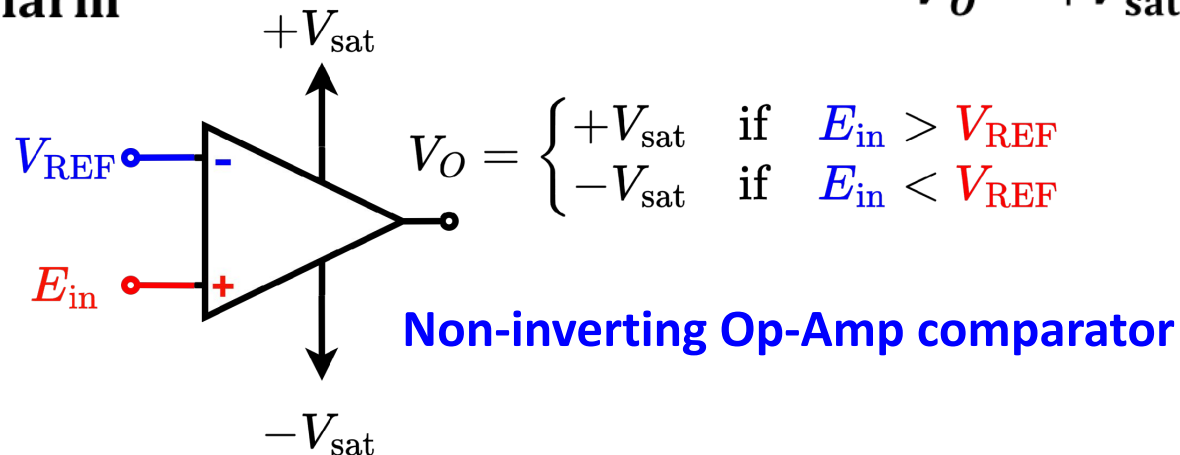
Smoke Detector - Operation

• No Smoke

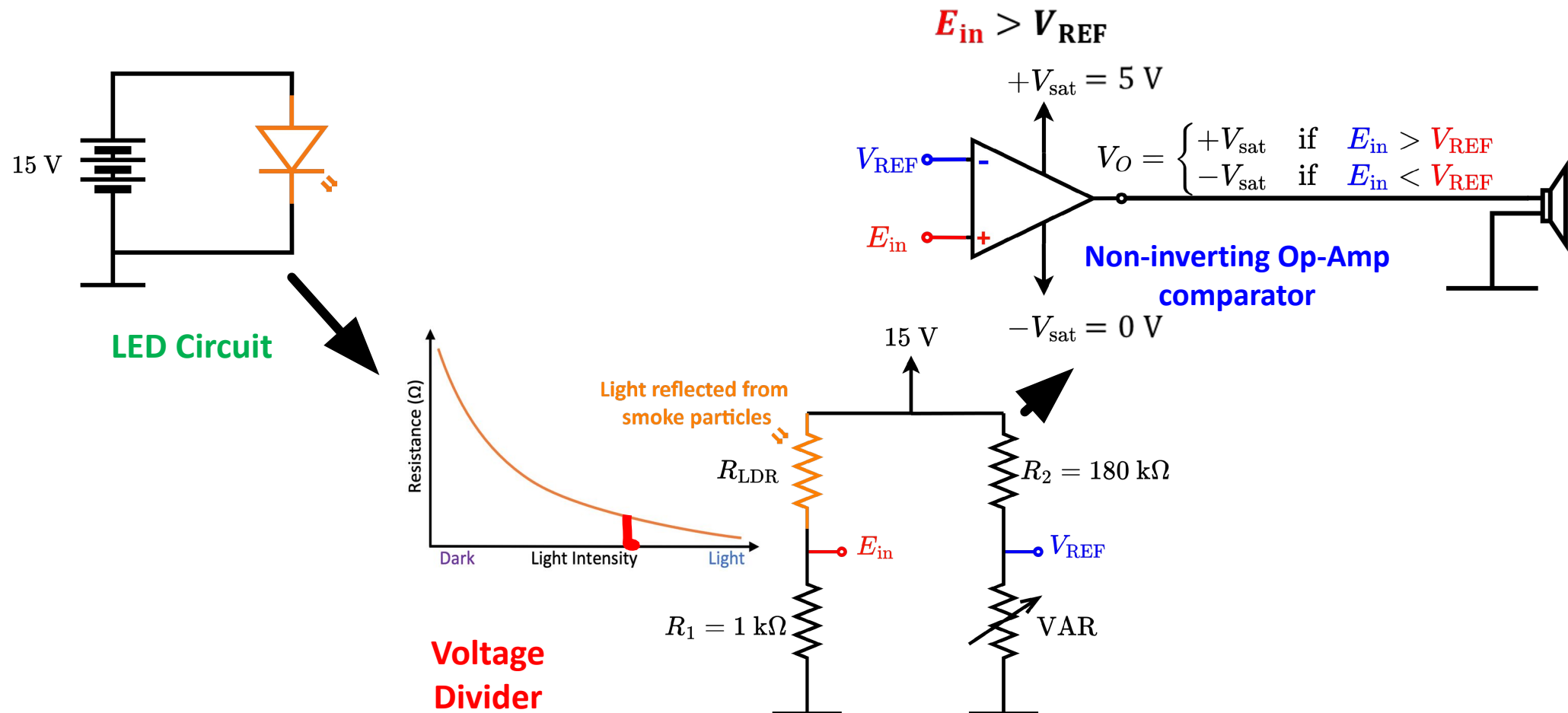
- No scattering
- Low intensity at LDR (Dark - $I \downarrow$)
- High resistance across LDR ($R_{LDR} \uparrow$)
- Decrease in E_{in}
- Possibility of ($E_{in} < V_{REF}$)
- $V_O = -V_{sat} \rightarrow$ No Alarm

• Smoke Present

- High scattering from smoke particles
- High intensity at LDR (Light - $I \uparrow$)
- Low resistance across LDR ($R_{LDR} \downarrow$)
- Increase in E_{in}
- Possibility of ($E_{in} > V_{REF}$)
- $V_O = +V_{sat} \rightarrow$ Alarm Rings!



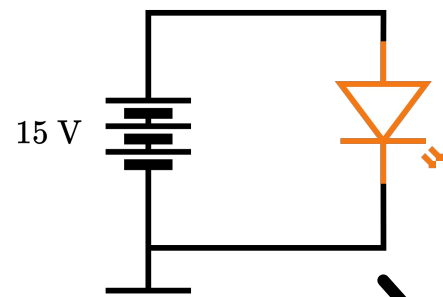
Smoke Detector - Operation



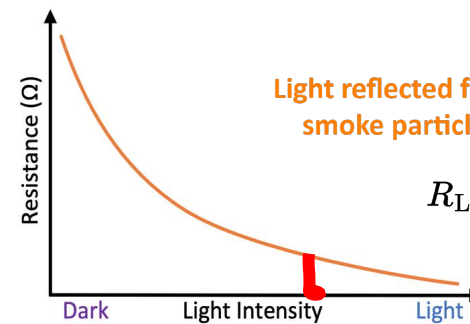
Smoke Detector - Operation

Scattering on to LDR → Bright - $I \uparrow \rightarrow R_{\text{LDR}} \downarrow \rightarrow E_{\text{in}} \uparrow \rightarrow$ **Positive saturation** $\rightarrow V_O = -V_{\text{sat}} \rightarrow$ **No Alarm**

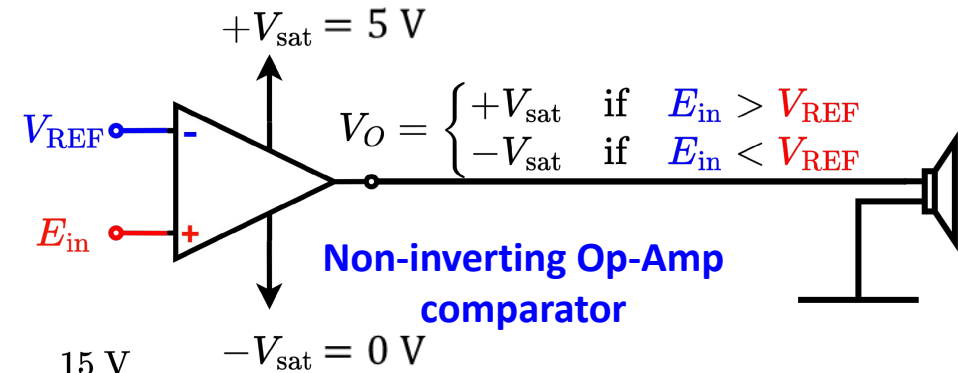
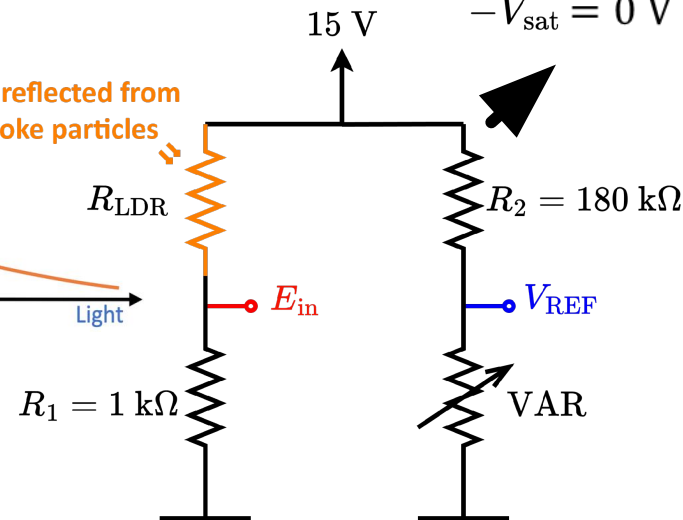
$$E_{\text{in}} > V_{\text{REF}}$$



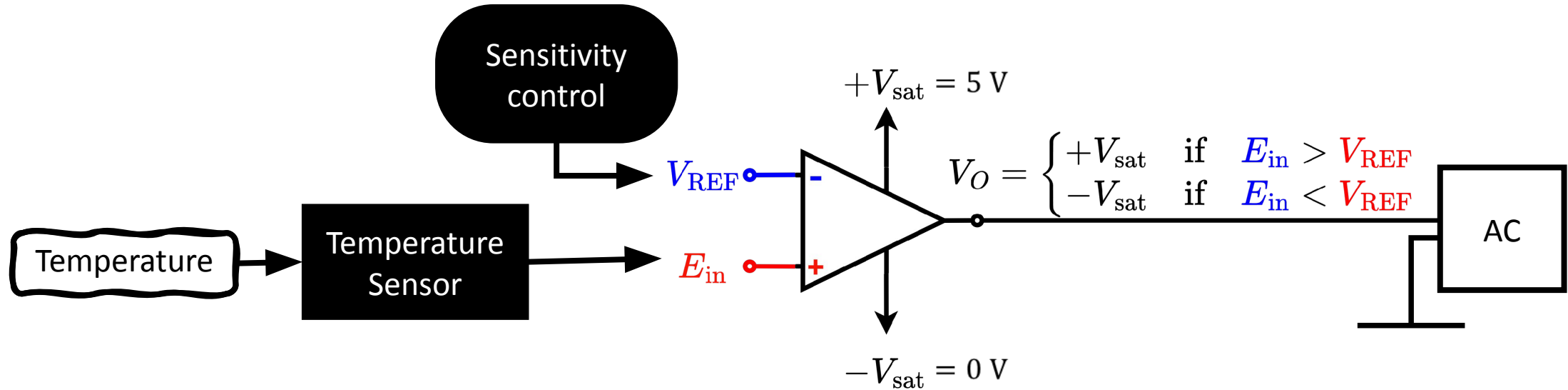
LED Circuit



Voltage Divider

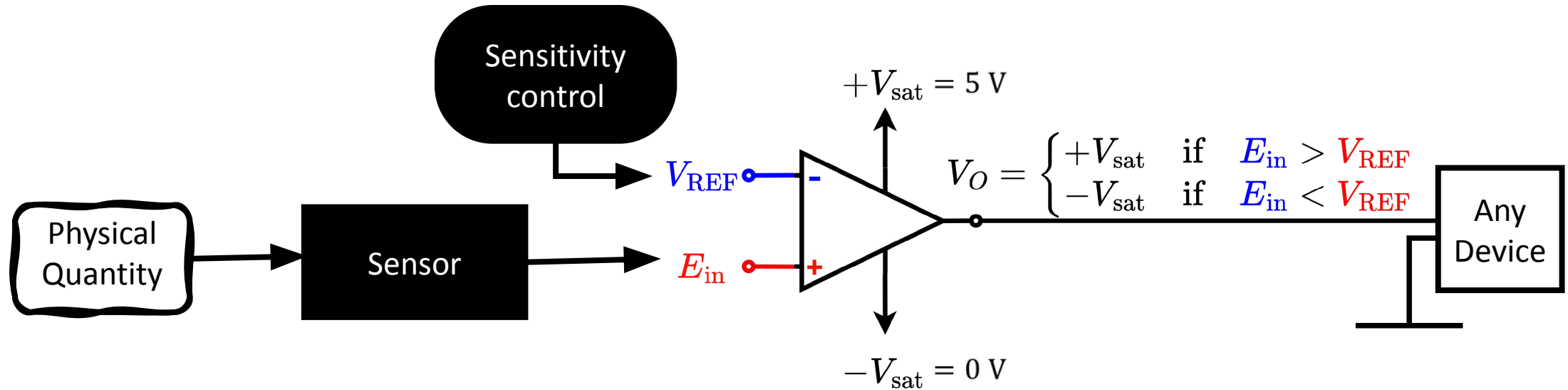


Comparator Application – Automatic AC



- Comparators can be used to switch on an **AC** based on a **Temperature**.
- The sensitivity of **switching on the AC** is determined by V_{REF}

Summary



- Comparators can be used to switch on **any device** based on a **physical quantity**.
- The sensitivity of **switching** is determined by V_{REF}

Thank you